Online Appendix

The Macroeconomic Effects of Global Supply Chain Shocks

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Contents

A Diagnostics of the Instrument **1**
A.1 Granger Causality and Correlation with Other Shocks . . . 1
A.2 Autocorrelation . . . 4
A.3 Instrument Strength . . . 5

B Robustness of Main Results **6**
B.1 Robustness to the COVID Period . . . 7
B.2 Invertibility Assumption . . . 9
B.3 Lag Length Sensitivity . . . 14
B.4 Additional Controls . . . 14
B.5 Estimation over the Identification Sample . . . 15

C The Role of Identification: A Recursive Benchmark **17**
C.1 Recursive Ordering Identification . . . 17
C.2 Comparison with the Main Results . . . 19

D Data **22**

E Price Surcharges & Narrative Analysis **25**
E.1 The Composition of Surcharge Announcements . . . 26
E.2 Step 1: Isolate Exogenous Supply-Side Events from Price Announcements . . . 28
E.3 Step 2: Corroborate Price Surcharges with External Sources . . . 35

References **91**

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A Diagnostics of the Instrument

This appendix presents a series of diagnostic checks on the FEU price surcharge series viewed as an external instrument for the global supply chain shock, following the approach outlined in Ramey (2016). The checks evaluate (i) its non-predictability from past macroeconomic conditions, (ii) its orthogonality to the leading structural shocks documented in the literature, (iii) the absence of serial correlation in the proxy, and (iv) the strength of the instrument from the first-stage regression.

A.1 Granger Causality and Correlation with Other Shocks

A desirable property is that the proxy should be orthogonal both to past macroeconomic conditions and to other identified structural shocks. If the instrument were predictable from lagged macroeconomic variables, it would no longer be exogenous to the rest of the system, and if it co-moved with shocks already identified in the literature, it would fail to isolate a global supply chain shock.

First, I examine whether the FEU price surcharge series can be forecasted using a broad set of macroeconomic variables. Table A.1 reports the results of Granger causality tests in which the FEU price surcharge series is regressed on its own twelve lags and two lags of each control variable. The p-values are all well above conventional significance thresholds, so the null hypothesis that the instrument is not Granger-caused by other variables cannot be rejected for any series. The same conclusion holds for the joint test reported in the last row of the table: the null that the lags of all controls are simultaneously zero is not rejected either, so the instrument is not systematically predictable from past macroeconomic indicators or from indices capturing global uncertainty, geopolitical risk, and US–China trade conditions, either one variable at a time or jointly.

I then assess whether the instrument is correlated with structural shock series commonly used in the literature. Table A.2 reports the Pearson correlation coefficients between the FEU

Table A.1: Granger Causality Tests

Series	Source	Transf	p-value	F-stat	df1	df2	Sample
World Oil Production	Baumeister and Hamilton (2019)	MoM	0.81	0.21	2	117	2014M8-2024M12
US Consumer Price Index	Haver Analytics	MoM	0.44	0.82	2	117	2014M8-2024M12
US Terms of Trade	Haver Analytics	Level	0.36	1.02	2	118	2014M8-2024M12
WTI Oil Price	Haver Analytics	MoM	0.86	0.15	2	117	2014M8-2024M12
US Industrial Production	Haver Analytics	MoM	0.68	0.38	2	117	2014M8-2024M12
US Unemployment Rate	Haver Analytics	Level	0.86	0.15	2	118	2014M8-2024M12
GSCPI	Benigno et al. (2022)	Level	0.43	0.86	2	118	2014M8-2024M12
World Uncertainty Index	Ahir, Bloom and Furceri (2022)	Level	0.62	0.48	2	118	2014M8-2024M12
US Trade Policy Uncertainty Index	Baker, Bloom and Davis (2016)	Level	0.48	0.74	2	118	2014M8-2024M12
World Sentiment Index	Ahir, Bloom and Furceri (2022)	Level	0.77	0.26	2	118	2014M8-2024M12
S&P500	Haver Analytics	MoM	0.72	0.33	2	117	2014M8-2024M12
Geopolitical Risk Index	Caldara and Iacoviello (2022)	Level	0.57	0.56	2	118	2014M8-2024M12
Global Economic Condition	Baumeister, Korobilis and Lee (2022)	Level	0.60	0.52	2	118	2014M8-2024M12
Oil Price Expectations 3Months	Baumeister (2023)	Level	0.95	0.05	2	98	2014M8-2023M4
US - China Tension Index	Rogers, Sun and Sun (2024)	Level	0.37	0.99	2	109	2014M8-2024M3
Federal Funds Rate	FRED	Level	0.87	0.13	2	118	2014M8-2024M12
Joint	–	–	0.21	1.26	32	67	2014M8-2023M4

Note: The table shows the results of a series of Granger Causality tests of the FEU price surcharge series using a selection of macroeconomic variables. The series are made stationary when necessary by taking the MoM growth rate. The tests are conducted by regressing the shock on its own 2 lags, 2 lags of the other variable and a constant. The test is on the joint significance of the lags of the additional variable. The last row (*Joint*) reports the test where the unrestricted model includes 2 lags of all variables simultaneously, and the null is that all the additional lags are jointly zero. * p<0.05, ** p<0.01, *** p<0.001.

Table A.2: Correlation with other Structural Shocks

Series	Source	ρ	p-value	t-stat	df	Sample
Oil News Shock	Känzig (2021)	-0.05	0.62	-0.50	123	2014M8-2024M12
Carbon News Shock	Känzig (2023)	0.11	0.37	0.91	63	2014M8-2019M12
Oil Supply Shock	Baumeister and Hamilton (2019)	-0.03	0.72	-0.36	123	2014M8-2024M12
Economic Activity Shock	Baumeister and Hamilton (2019)	0.02	0.86	0.18	123	2014M8-2024M12
Oil Consumption Demand Shock	Baumeister and Hamilton (2019)	-0.14	0.12	-1.56	123	2014M8-2024M12
Oil Inventory Demand Shock	Baumeister and Hamilton (2019)	0.01	0.91	0.12	123	2014M8-2024M12
Monetary Policy Shock	Bauer and Swanson (2023)	-0.05	0.61	-0.51	111	2014M8-2023M12

Note: The table shows the correlation of the FEU price surcharge series with a wide range of structural shocks from the literature. ρ is the Pearson correlation coefficient, the p-value corresponds to the test whether the correlation is different from zero, t-stat is the corresponding test statistic, df are the degrees of freedom. * p<0.05, ** p<0.01, *** p<0.001.

price surcharge series and a wide range of structural shocks, including oil demand and supply shocks, monetary policy shocks, and other shocks. None of the correlations is statistically significant. The instrument is therefore uncorrelated with the catalogue of structural shocks already identified in the literature, so the variation it captures is not a relabelling of any of them.

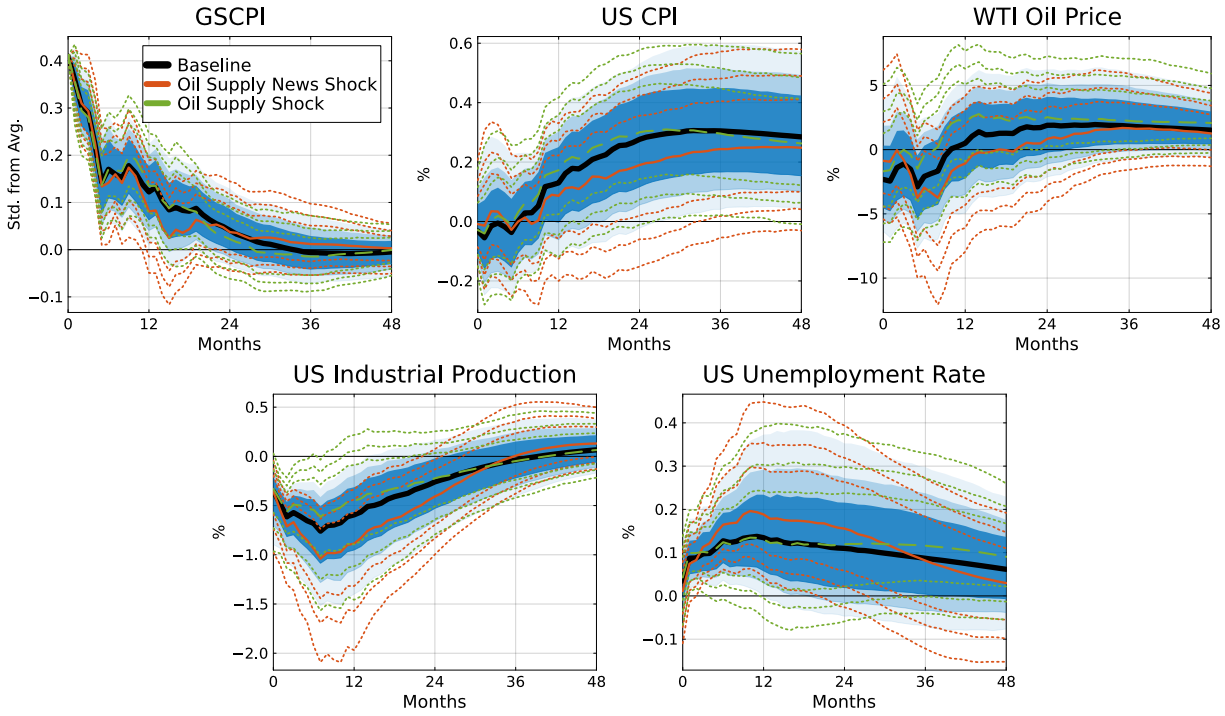


Figure A.1: Robustness to Controlling for Identified Oil Shocks

Note: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The black line with blue shaded bands is the baseline specification; the orange line is the specification that adds the oil supply news shock of Känzig (2021), and the green line the specification that adds the oil supply shock of Baumeister and Hamilton (2019), each with dotted confidence bands. The bands denote 68, 80 and 90 percent confidence intervals.

A related concern, raised in particular for the events at the Strait of Hormuz, is that some disruptions could reflect oil-market shocks rather than shocks to the global supply chain. The correlation results above already speak against this, since the instrument is uncorrelated with identified oil supply and oil demand shocks. As a more direct check, I augment the baseline VAR with an identified oil shock as an additional variable, keeping the WTI oil price in the system, and re-estimate the model with one oil shock at a time: the oil supply

news shock of Känzig (2021) and the oil supply shock of Baumeister and Hamilton (2019). Figure A.1 reports the resulting impulse responses. Under both specifications the responses lie essentially on top of the baseline, confirming that the identified global supply chain shock is not confounded by oil-market disturbances.

Taken together, the diagnostic results in this section support the interpretation of the FEU price surcharge series as a valid external instrument: it is not forecastable from past macroeconomic variables either individually or jointly, and it is orthogonal to other structural shocks commonly used in the literature, reinforcing the exogeneity condition required for Proxy-SVAR identification.

A.2 Autocorrelation

Any serial correlation in the proxy would imply that current realizations are predictable from past values, which would be inconsistent with its role of capturing only the unanticipated component of a structural shock and would contaminate the impulse responses with anticipation effects.

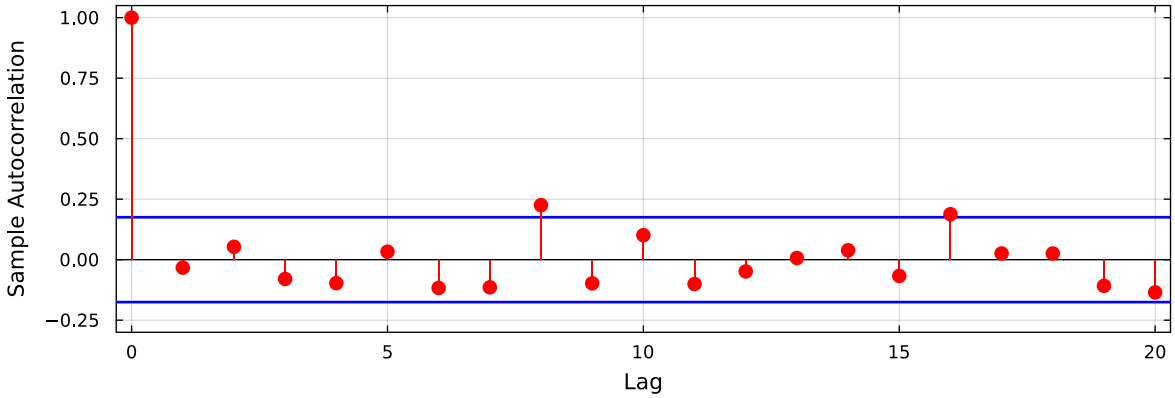


Figure A.2: The Autocorrelation Function of the FEU Price Surcharge Series

Figure A.2 plots the autocorrelation function of the FEU price surcharge series; the autocorrelations are statistically indistinguishable from zero at all lags except for the 8th lag, which lies slightly outside the confidence band. Table A.3 complements the figure with Ljung-

Box Q-statistics testing the joint null that the autocorrelations up to a given lag are zero, evaluated at lags 6 through 20. The associated p-values are uniformly above conventional significance thresholds, so the joint null cannot be rejected at any horizon. Taken together, the autocorrelogram and the Ljung-Box tests provide no evidence of serial correlation in the FEU price surcharge series, so the proxy satisfies the no-autocorrelation requirement expected of a valid Proxy-SVAR instrument.

Table A.3: Ljung–Box Test on the FEU Price Surcharge Series

Lag	6	8	10	12	14	16	18	20
Q-statistic	4.53	13.18	15.90	17.64	17.86	23.65	23.85	28.35
p-value	0.61	0.11	0.10	0.13	0.21	0.10	0.16	0.10

Note: The table reports the Ljung-Box Q-statistic and the corresponding p-value for the FEU price surcharge proxy at different lags. The null hypothesis is that the autocorrelations of the proxy are jointly zero up to the specified lag, with $Q \sim \chi^2(\text{lag})$. Sample: 2014M8-2024M12 ($n = 125$). * p<0.05, ** p<0.01, *** p<0.001.

A.3 Instrument Strength

To assess instrument relevance I report first-stage diagnostics for the Proxy-SVAR. Because the COVID-19 period induces a sharp variance break in the reduced-form residuals that materially contaminates conventional first-stage statistics, I estimate the first-stage regression jointly with the reduced-form VAR using the feasible-GLS procedure adapted from Lenza and Primiceri (2022) and Hamilton (Forthcoming), which downweights observations in the COVID regime by an estimated variance scaling factor $\hat{\kappa}^2$. The first-stage F-statistic and the heteroskedasticity-robust F-statistic are therefore computed on residuals that already control for the largest source of heteroskedasticity in the sample. Table A.4 reports both statistics for the baseline FEU price surcharge series and for the leave-one-category-out variants used in the robustness analysis.

Table A.4: First-Stage Regressions (Feasible GLS)

Specification	**Coef.**	F**-stat**	**Robust** F**-stat**	R^2	**Adj.**R^2	**Obs.**	**% Baseline**
Baseline	0.4082	13.21	12.40	9.70%	8.97%	43	100
IV Excl. Operational	0.3067	4.53	3.08	3.82%	2.98%	30	70
IV Excl. War	0.4605	8.07	17.87	6.16%	5.40%	36	84
IV Excl. Strike	0.4199	13.81	13.06	10.09%	9.36%	38	88
IV Excl. Weather	0.4249	12.96	11.94	9.54%	8.80%	32	74
IV Indicator	140.4590	6.89	5.88	5.31%	4.54%	43	100
IV Excl. US ports	0.3937	11.82	10.85	9.03%	8.29%	41	95

Note: The table reports first-stage statistics from the regression of the GSCPI VAR residual on the supply chain disruption instrument, across alternative treatments of the COVID-19 period. The GLS specification follows Lenza and Primiceri (2022) and Hamilton (Forthcoming): COVID observations (Jan.–Dec. 2020) are downweighted by the estimated variance scaling factor $\hat{\kappa}^2$. "Obs." is the number of months with a non-zero realization of the instrument; % Baseline is the percentage of non-zero realizations relative to the baseline instrument.

The baseline instrument has $F = 13.21$ and a robust $F = 12.40$, both well above the conventional threshold of 10, so the FEU price surcharge series is highly relevant. The strength is preserved when individual disruption categories are excluded, with the single exception of the no-operational variant: dropping operational disruptions removes removes roughly 40% of the events in the proxy (about 30% of the non-zero monthly observations), and several of those discarded events capture disruptions of unusually broad scope, affecting multiple trade lanes or critical chokepoints; prominent examples include the 2021 closure of the Suez Canal and the 2021 closure of the Meishan Terminal at Ningbo, one of the largest container ports in the world. Once these events are removed, the operational-free instrument is mechanically much weaker, but the strength of the baseline FEU surcharge series used throughout the paper is unaffected.

B Robustness of Main Results

To ensure the reliability of the baseline results, I perform a comprehensive set of robustness checks. The first two address concerns about the role of the COVID-19 period and the

validity of the invertibility assumption. The next two examine the sensitivity of the impulse responses to the dynamic structure of the VAR, namely the choice of lag length and the inclusion of additional control variables.

B.1 Robustness to the COVID Period

The COVID-19 period raises two concerns: (i) the elevated volatility of the pandemic and post-pandemic years could distort the estimation of the reduced-form VAR, and (ii) the proxy itself could be contaminated by events whose proximate cause is the pandemic rather than a global supply chain disruption. The figures in this subsection address the first concern; the narrative cross-validation in Appendix E and the no-COVID-events specification reported in Section 6 Figure 13 of the paper address the second.

Dropping the 2020 calendar year.— As a first check, I re-estimate the model after dropping the entire 2020 calendar year from the sample. Figure B.3 reports the resulting impulse responses, which are essentially indistinguishable from the baseline at every horizon and across all five variables. Even removing the COVID period entirely from the estimation does not alter the conclusions of the paper.

Extending the GLS scaling block to 2020–2024.— The baseline specification downweights only the 2020 observations through the variance scaling factor $\hat{\kappa}^2$. As a second, more aggressive check, I extend the GLS scaling block to cover the full 2020–2024 window, downweighting both the pandemic and the post-pandemic observations in the VAR estimation as well as in the construction of the instrument. Figure B.4 shows the resulting impulse responses. The main findings are confirmed: a global supply chain shock generates a persistent disruption to the global supply chain and stagflationary effects on consumer prices and industrial production whose magnitudes and shapes are comparable to the baseline, while the responses of unemployment and the WTI oil price remain statistically insignificant.

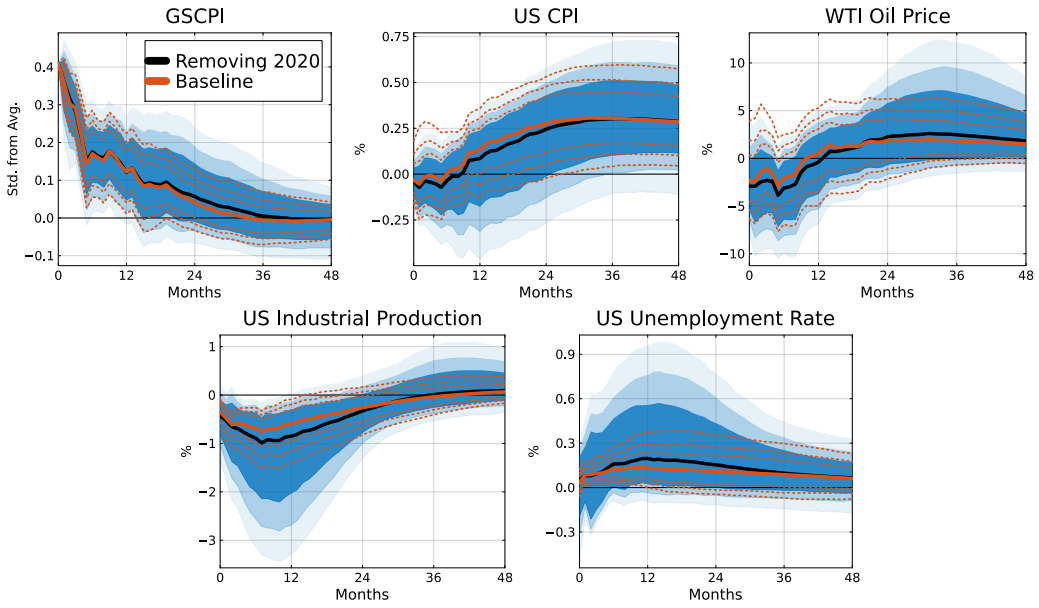


Figure B.3: Robustness to Dropping the 2020 Calendar Year

Note: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The orange lines represent the baseline specification; the black lines represent the estimates obtained when the entire 2020 calendar year is dropped from the sample. The solid lines are point estimates and the shaded areas (and dotted lines) denote 68, 80 and 90 percent confidence bands.

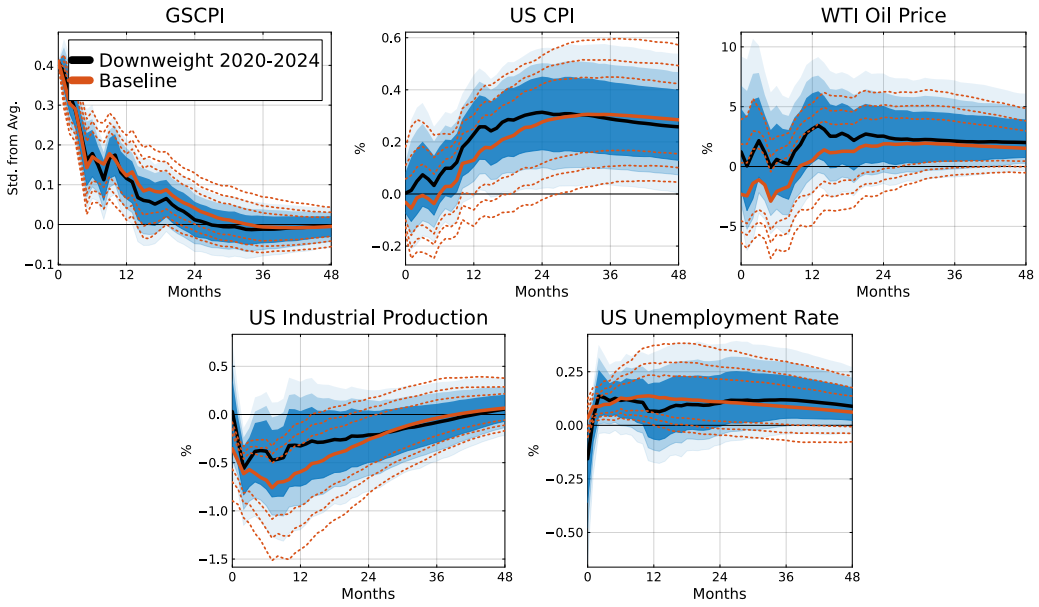


Figure B.4: Robustness to a Full 2020–2024 GLS Scaling Block

Note: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The orange lines represent the baseline specification; the black lines represent the estimates obtained when the GLS scaling is extended to the full 2020–2024 window. The solid lines are point estimates and the shaded areas (and dotted lines) denote 68, 80 and 90 percent confidence bands.

B.2 Invertibility Assumption

The identification strategy requires the structural shocks to lie within the linear span of current and lagged values of the endogenous variables. I provide two complementary checks: a direct test of the invertibility assumption based on lagged values of the instrument added to the VAR equations, and an invertibility-robust estimator that does not require the assumption to hold.

Augmented-VAR test of invertibility.— For each variable i in the baseline VAR, I follow the procedure of Plagborg-Møller and Wolf (2022) and augment the corresponding equation with m lags of the instrument:

$$y_{it} \;=\; \pi_i' \boldsymbol{x}_{t-1} \;+\; \alpha_1 z_{t-1} \;+\; \alpha_2 z_{t-2} \;+\; \cdots \;+\; \alpha_m z_{t-m} \;+\; \eta_{it}, \tag{1}$$

where y_{it} denotes the i-th VAR variable, $\boldsymbol{x}_{t-1}$ collects the baseline VAR regressors (constant and twelve own-lags of all five endogenous variables), and z_t is the FEU price surcharge series. Under the null of invertibility, lagged values of the instrument should add no predictive content to the VAR equations once the endogenous regressors are included, that is, $H_0 : \alpha_1 = \cdots = \alpha_m = 0$ for every equation i. The first panel of Table B.5 reports the p-values from the equation-by-equation F-tests for $m \in \{6, 7, \ldots, 12\}$. Across all five series and all lag lengths, the p-values are above conventional significance thresholds, so the coefficients on the lagged instrument are statistically indistinguishable from zero in every equation. The second panel reports the joint test that pools the lags of the instrument across all five equations simultaneously, following the joint variant of Hamilton (Forthcoming): this also fails to reject the joint null at $m = 12$, with a p-value of 0.99. The data are therefore consistent with invertibility, both equation by equation and jointly across the system.

Invertibility-robust proxy-SVAR.— I also estimate the impulse responses with the invertibility-robust proxy-SVAR procedure of Forni, Gambetti and Ricco (2022), which does not require the structural shocks to lie within the linear span of the VAR variables. The procedure

Table B.5: Testing the invertibility assumption

Lags	US CPI	WTI Oil Price	US Ind Prod	US Unemployment	GSCPI
6	0.12	0.41	0.96	0.69	0.80
7	0.17	0.15	0.96	0.74	0.88
8	0.17	0.16	0.98	0.79	0.74
9	0.18	0.19	0.99	0.86	0.58
10	0.24	0.24	0.95	0.84	0.36
11	0.32	0.30	0.92	0.89	0.40
12	0.30	0.27	0.95	0.93	0.49

Joint test: 12 lags p-value = 0.99 F-stat = 0.54 $df_1 = 60$ $df_2 = 200$

Note: The first panel shows the p-values of a series of F-test that the coefficients $\alpha_1, \ldots, \alpha_m$ are zero in the regression $y_{it} = \pi_i' \mathbf{x}_{t-1} + \alpha_1 z_{t-1} + \cdots + \alpha_m z_{t-m} + \eta_{it}$. The test is conducted for each series included in the baseline VAR model, for different numbers of lags of the instrument. The lag order of the VAR is set to 12 and in terms of deterministics, only a constant is included. The second panel shows the joint system test across all i that lags of z do not appear in any of the equations. * p<0.05, ** p<0.01, *** p<0.001.

operates in two steps. First, the proxy is cleaned of any predictable component generated by past values of itself and of the endogenous VAR variables through the regression

$$\tilde{z}_t \;=\; c \;+\; \beta(L)\,\tilde{z}_{t-1} \;+\; \mu(L)'\,\boldsymbol{x}_{t-1} \;+\; z_t, \tag{2}$$

where $\beta(L)$ collects the coefficients on m lags of the instrument $\tilde{z}_t$ and $\mu(L)$ those on q lags of the endogenous variables $\boldsymbol{x}_{t-1}$, taken to be the same five variables that enter the baseline VAR (US CPI, WTI oil price, US industrial production, US unemployment rate, and the GSCPI). Second, the cleaned proxy z_t is projected on the contemporaneous and r lead reduced-form VAR residuals,

$$z_t \;=\; \delta_0'\,\hat{\boldsymbol{\varepsilon}}_t \;+\; \delta_1'\,\hat{\boldsymbol{\varepsilon}}_{t+1} \;+\; \cdots \;+\; \delta_r'\,\hat{\boldsymbol{\varepsilon}}_{t+r} \;+\; v_t, \tag{3}$$

and the structural impact effects of interest are estimated using the fitted values $\hat{z}_t = \hat{\delta}(F)\varepsilon_t$. Figure B.5 reports the resulting impulse responses for 216 different specifications of the procedure, obtained by varying the three parameters m, q and r over a grid. All the responses lie tightly on top of the baseline Proxy-SVAR responses, regardless of the parameterization chosen.

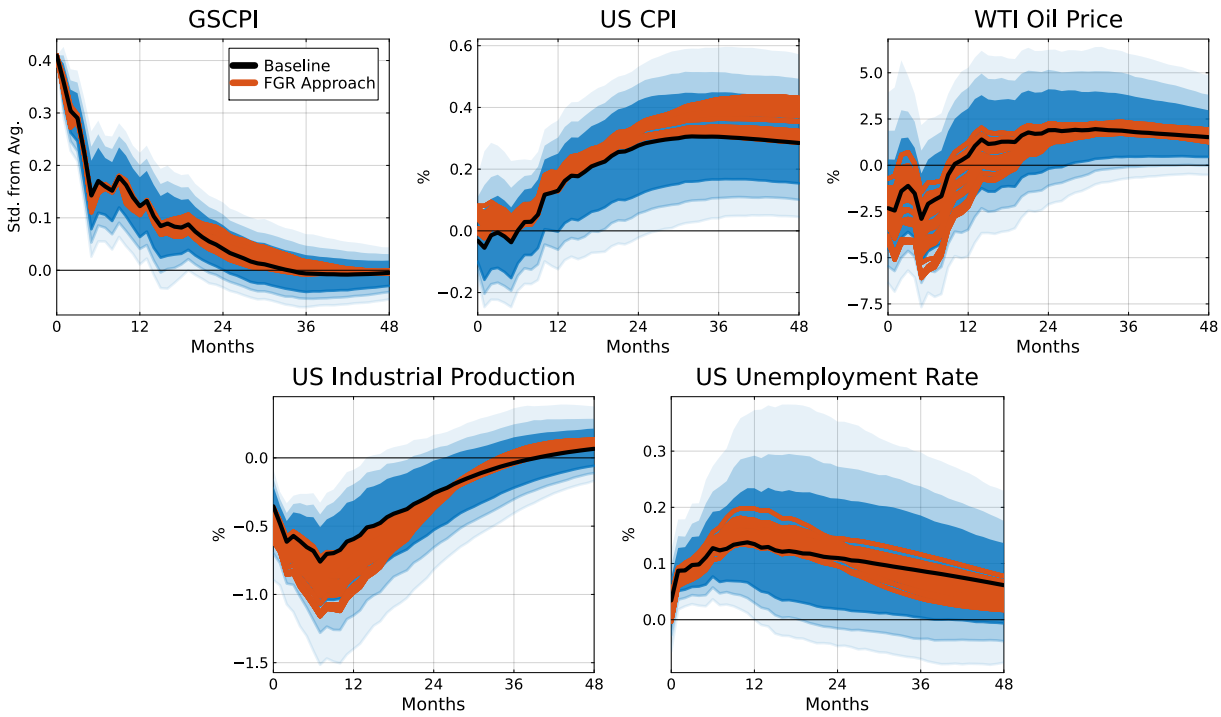


Figure B.5: Invertibility-Robust Impulse Responses

Note: Impulse responses to a global supply chain shock corresponding to a $1066 price surcharge. The black line is the baseline Proxy-SVAR point estimate, surrounded by 68, 80, and 90 percent confidence bands. The orange lines report the impulse responses obtained by varying the parameters $m \in \{1, \ldots, 6\}$ (lags of the instrument in the cleaning regression), $q \in \{1, \ldots, 6\}$ (lags of the VAR variables in the cleaning regression), and $r \in \{1, \ldots, 6\}$ (leads of the reduced-form residuals in the lead regression).

To understand why the invertibility-robust impulse responses are essentially identical to the baseline, two diagnostic tests can be run on the underlying regressions. The first is the joint null

$$H_0^{(1)} : \beta_1 = \cdots = \beta_m = 0 \tag{4}$$

in the cleaning regression (2): failing to reject indicates that the raw proxy is already or-

thogonal to its own past and to the lagged VAR variables, so the cleaning step is empty. The second is the joint null

$$H_0^{(2)} : \delta_1 = \cdots = \delta_r = 0 \tag{5}$$

in the lead regression (3): failing to reject is the formal counterpart of the invertibility condition for an external proxy.

Table B.6 reports the p-values of both tests for $m \in \{1, \ldots, 12\}$ in Panel A and $r \in \{1, \ldots, 6\}$ in Panel B. None of the rows in either panel rejects at conventional significance levels: the p-values lie uniformly above 0.10, with most well above 0.20. Because the cleaning regression coefficients $\beta(L)$ are jointly indistinguishable from zero, the cleaned proxy coincides essentially with the raw FEU price surcharge series; and because the lead coefficients $\delta_1, \ldots, \delta_r$ are jointly indistinguishable from zero, no bias correction is induced by relaxing invertibility. The two diagnostics therefore explain why the invertibility-robust impulse responses in Figure B.5 overlap the baseline: every parameter that would otherwise differentiate the two specifications is statistically zero.

Table B.6: Diagnostic Tests for the Invertibility-Robust Estimator

lags/leads	p-value	F-Statistic	DF1	DF2
Panel A: Testing the lags				
1	0.30	1.09	1	87
2	0.58	0.54	2	86
3	0.48	0.83	3	85
4	0.42	0.99	4	84
5	0.53	0.83	5	83
6	0.42	1.02	6	82
7	0.20	1.44	7	80
8	0.15	1.57	8	78
9	0.17	1.49	9	76
10	0.20	1.39	10	74
11	0.22	1.34	11	72
12	0.23	1.32	12	70
Panel B: Testing invertibility				
1	0.20	1.49	5	113
2	0.40	1.06	10	107
3	0.39	1.07	15	101
4	0.38	1.08	20	95
5	0.49	0.99	25	89
6	0.70	0.84	30	83

Note: **Panel A** reports F-tests of the null $H_0 : \beta_1 = \cdots = \beta_m = 0$ in the cleaning regression $\tilde{z}_t = c + \beta(L)\tilde{z}_{t-1} + \mu'(L)\mathbf{x}_{t-1} + w_t$, where $\mathbf{x}_{t-1}$ collects the lagged VAR variables ($p = 6$ lags, with MoM growth rates (%) for US CPI, WTI Oil Price, US Industrial Production, US Unemployment Rate, GSCPI) and the test is conducted varying the number of instrument lags $m \in \{1, \ldots, 12\}$. **Panel B** reports F-tests of the null $H_0 : \delta_1 = \cdots = \delta_r = 0$ in the lead regression $\tilde{z}_t = \delta_0'\hat{\varepsilon}_t + \delta_1'\hat{\varepsilon}_{t+1} + \cdots + \delta_r'\hat{\varepsilon}_{t+r} + v_t$, where $\hat{\varepsilon}_t$ are the residuals of a VAR($p = 12$) and the test is conducted varying the number of leads $r \in \{1, \ldots, 6\}$. * p<0.05, ** p<0.01, *** p<0.001.

B.3 Lag Length Sensitivity

I assess the sensitivity of the aggregate IRFs to alternative lag lengths. Figure B.6 displays structural impulse responses estimated from VAR specifications including 14, 16, 18, and 20 lags, in addition to the 12-lag baseline. The results are robust across specifications, with the dynamic causal effects of the global supply chain shock remaining statistically and economically consistent regardless of lag length. One exception is the WTI oil price series, which exhibits notable variation across specifications; however, these responses are not statistically significant and have wide confidence intervals. The main takeaway is that the results are stable across lag structures and consistent with the baseline specification.

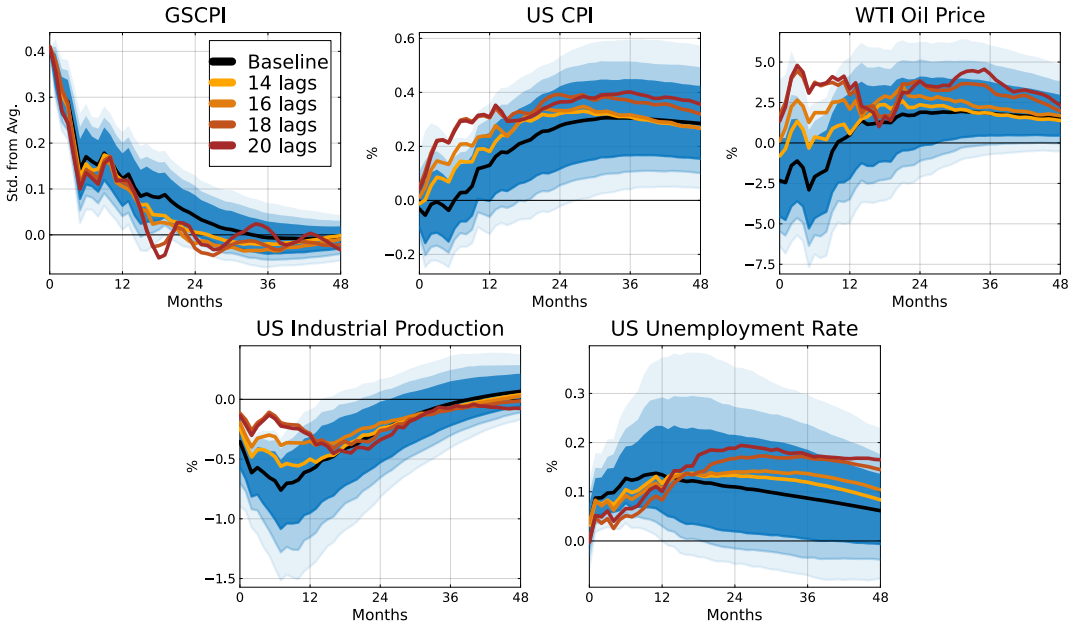


Figure B.6: Robustness with respect to Lag Length

Note: Impulse responses to a global supply chain shock corresponding to a $1,066 price surcharge. The solid lines are the point estimates and the shaded areas are 68, 80 and 90 percent confidence bands for the baseline model, respectively.

B.4 Additional Controls

I examine whether the impulse responses are robust to the inclusion of additional control variables. Specifically, I estimate alternative specifications of the baseline VAR by sequentially

adding one control variable at a time from a broad set of economic indicators, including price expectations, sectoral price and production indices, services, world output, interest rates, and other macro-financial variables. The full list of variables, totalling 73 additional controls, is documented in Appendix D. Figure B.7 shows that the estimated impulse responses remain remarkably stable across these specifications. These results demonstrate that the dynamic causal effects of global supply chain shocks are robust not only to lag structure but also to the inclusion of a wide range of controls, reinforcing the credibility of the baseline findings.

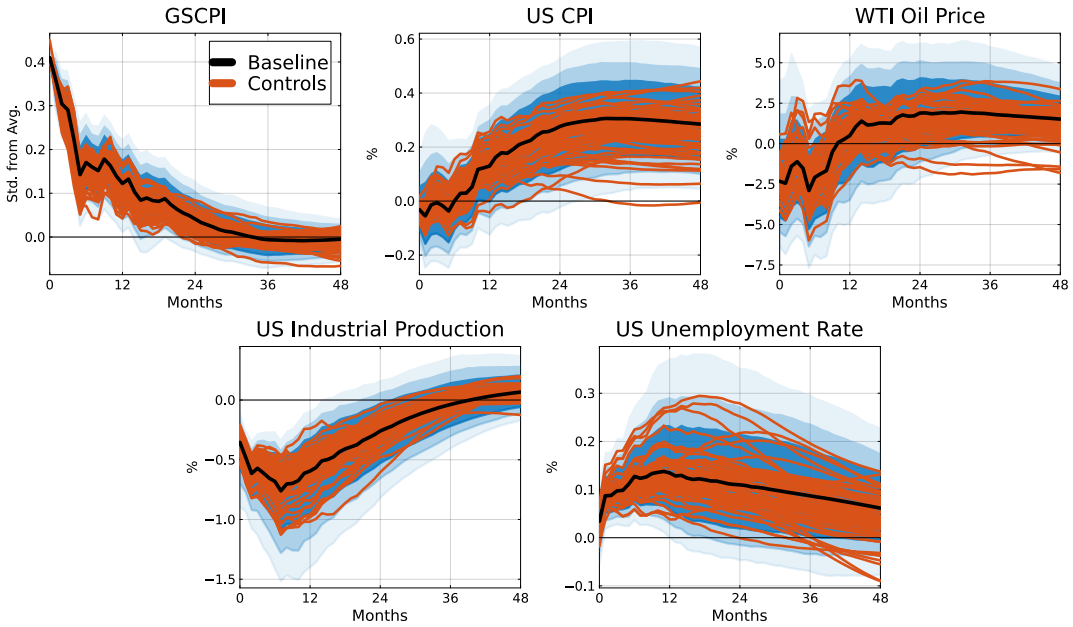


Figure B.7: Robustness with respect to Additional Controls

Note: Impulse responses to a global supply chain shock corresponding to a $1,066 price surcharge. The solid black line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands for the baseline model. The solid orange lines are the point estimates for the model with one additional control included at a time.

B.5 Estimation over the Identification Sample

The baseline VAR is estimated over the sample 1998–2024, while the FEU price surcharge instrument covers 2014–2024. Within the Proxy-SVAR framework, the identification of the structural impact vector requires only the reduced-form innovations over the coverage of the instrument, whereas the autoregressive coefficients that govern the propagation of the shock

are estimated using information over the full sample. As an additional robustness check, I estimate the VAR autoregressive coefficients starting in 2013, so that, after the twelve lags, the effective estimation sample coincides with the identification sample. Figure B.8 compares the resulting impulse responses with the baseline. The responses are similar in shape and magnitude, though less precisely estimated given the shorter sample. The baseline estimates lie well within the confidence bands of the short-sample model, indicating that the baseline results are not an artifact of the longer estimation sample.

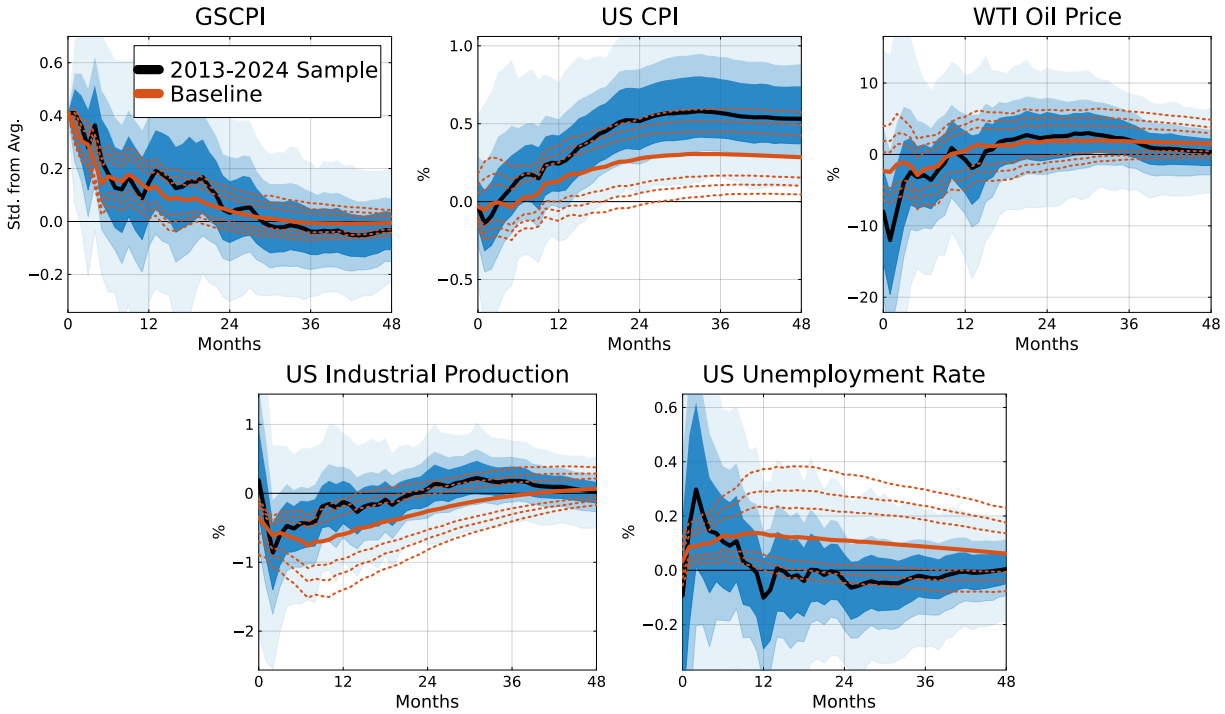


Figure B.8: Robustness to Estimation over the Identification Sample

Note: Impulse responses to a global supply chain shock corresponding to a $1,066 price surcharge. The solid black lines are the point estimates from the VAR estimated over 2013–2024, with shaded areas denoting the 68, 80 and 90 percent confidence bands; the solid orange lines are the baseline point estimates, with dotted orange lines the corresponding confidence bands.

C The Role of Identification: A Recursive Benchmark

Indicators of global supply chain pressure, such as delivery times, port congestion, and shipping costs, are endogenously determined: they respond to fluctuations in global demand and production activity as much as to genuine disruptions on the supply side. This endogeneity is the central identification challenge of the paper, and the reason why an external instrument, built exclusively from narratively verified exogenous supply-side events, is needed to isolate the supply component of supply chain pressure. In this appendix, I estimate the same VAR as in the main text and identify the global supply chain shock through a recursive ordering, and compare the resulting impulse responses, forecast error variance decomposition, and historical decompositions with the baseline results. If the two identifications delivered similar answers, endogeneity concerns would be second-order in practice and the narrative instrument superfluous. The differences documented below are instead large and systematic. Under the recursive ordering, global supply chain shocks have essentially no real effects: production and unemployment do not respond, the shocks contribute almost nothing to the pandemic-era production shortfall, and they do not explain business-cycle fluctuations in real activity. More fundamentally, this pattern suggests that the shock identified by the recursive ordering may blend genuine supply disruptions with demand-driven pressure, a contamination that threatens its structural interpretation.

C.1 Recursive Ordering Identification

The reduced-form VAR is identical to the baseline model: the same five variables, twelve lags, the same sample, and feasible GLS treatment of the COVID observations. Recall from the main text the reduced-form representation and the invertibility assumption,

$$\boldsymbol{y}_t = \boldsymbol{\Pi}\, \boldsymbol{x}_{t-1} + \boldsymbol{\epsilon}_t, \qquad\qquad \boldsymbol{\epsilon}_t = \boldsymbol{A}^{-1}\, \boldsymbol{u}_t, \tag{6}$$

where $\boldsymbol{x}_{t-1}$ collects an intercept and p lags of the variables, $\boldsymbol{\epsilon}_t \sim (\mathbf{0}, \boldsymbol{\Omega})$ is the vector of reduced-form innovations, $\boldsymbol{A^{-1}}$ is the n x n structural impact matrix, and $\boldsymbol{u}_t \sim (\mathbf{0}, \boldsymbol{D})$ is the vector of mutually uncorrelated structural shocks, with $\boldsymbol{D}$ diagonal. Identification is achieved by imposing that $\boldsymbol{A^{-1}}$ is lower triangular and that the structural shocks have unit variance, $\boldsymbol{D} = \boldsymbol{I}_n$. Under these identifying assumptions, $\boldsymbol{A^{-1}}$ is uniquely pinned down as the Cholesky factor of the reduced-form variance-covariance matrix, $\boldsymbol{\Omega} = \boldsymbol{A^{-1}}(\boldsymbol{A^{-1}})'$. Since the VAR is stationary, (6) yields a VMA(∞) representation of the form

$$\boldsymbol{y}_t = \sum_{h=0}^{\infty} \boldsymbol{\Psi}_h \boldsymbol{\epsilon}_{t-h}, \tag{7}$$

where the weights $\boldsymbol{\Psi}_h$ are functions of the reduced-form coefficient matrix $\boldsymbol{\Pi}$.

With $\boldsymbol{A^{-1}}$ in hand, all the objects of interest are identified. The vector of structural impulse responses at horizon h to the shock in the j-th position of the system is

$$\boldsymbol{\theta}_h^j = \boldsymbol{\Psi}_h \, \boldsymbol{h}_j, \qquad \boldsymbol{h}_j = \boldsymbol{A^{-1}} \boldsymbol{e}_j, \tag{8}$$

where $\boldsymbol{e}_j$ is the j-th unit column vector and $\boldsymbol{h}_j$ is the impact vector of the shock. At each horizon h, (8) traces the response of every variable to the shock. The historical decomposition cumulates these responses over the realized history of the shock. The structural shocks are recovered by inverting the impact matrix,

$$\boldsymbol{u}_t = \boldsymbol{A} \, \boldsymbol{\epsilon}_t, \tag{9}$$

and the contribution of the j-th shock, u_{jt}, to the observed path of the variables is

$$\boldsymbol{y}_t(u_j) = \sum_{h=0}^{t-1} \boldsymbol{\theta}_h^j \, u_{jt-h}, \tag{10}$$

where t indexes the observations from the start of the decomposition window. Lastly, I

compute the contribution of the global supply chain shock to the forecast error variance-covariance matrix of $\boldsymbol{y}_t$ at horizon h as

$$\sum_{s=0}^{h} \boldsymbol{\theta}_s^j \, (\boldsymbol{\theta}_s^j)'. \tag{11}$$

The ratio of the diagonal elements of (11) to those of the overall forecast error variance-covariance matrix at horizon h, $\sum_{s=0}^{h} \boldsymbol{\Psi}_s \boldsymbol{\Omega} \boldsymbol{\Psi}_s'$, provides the percentage of the variance of $\boldsymbol{y}_t$ at horizon h accounted for by the global supply chain shock, with horizon zero corresponding to the impact period. Impulse responses are rescaled so that the shock raises the GSCPI on impact by the same amount as in the baseline, a scale convention none of the results depends on, and confidence bands are obtained from the a moving-block bootstrap with 4,000 replications.

Following Benigno et al. (2022), I place the GSCPI after the macroeconomic aggregates and before the price of oil:

$$\boldsymbol{y}_t = (\text{CPI}_t,\ \text{IP}_t,\ \text{U}_t,\ \text{GSCPI}_t,\ \text{Oil}_t)'. \tag{12}$$

The GSCPI thus responds contemporaneously to innovations in consumer prices, production, and unemployment but not to oil, and the macroeconomic aggregates react to the global supply-chain shock only with a one-month lag.

C.2 Comparison with the Main Results

Figure C.9 compares the impulse responses under the two identifications; by construction, both shocks raise the GSCPI by the same amount on impact. The price responses are similar. The responses of industrial production and unemployment, by contrast, are small and not statistically significant.

A plausible explanation is the contamination of demand. Demand shocks move output and prices in the same direction, while supply disruptions move them in opposite directions, so

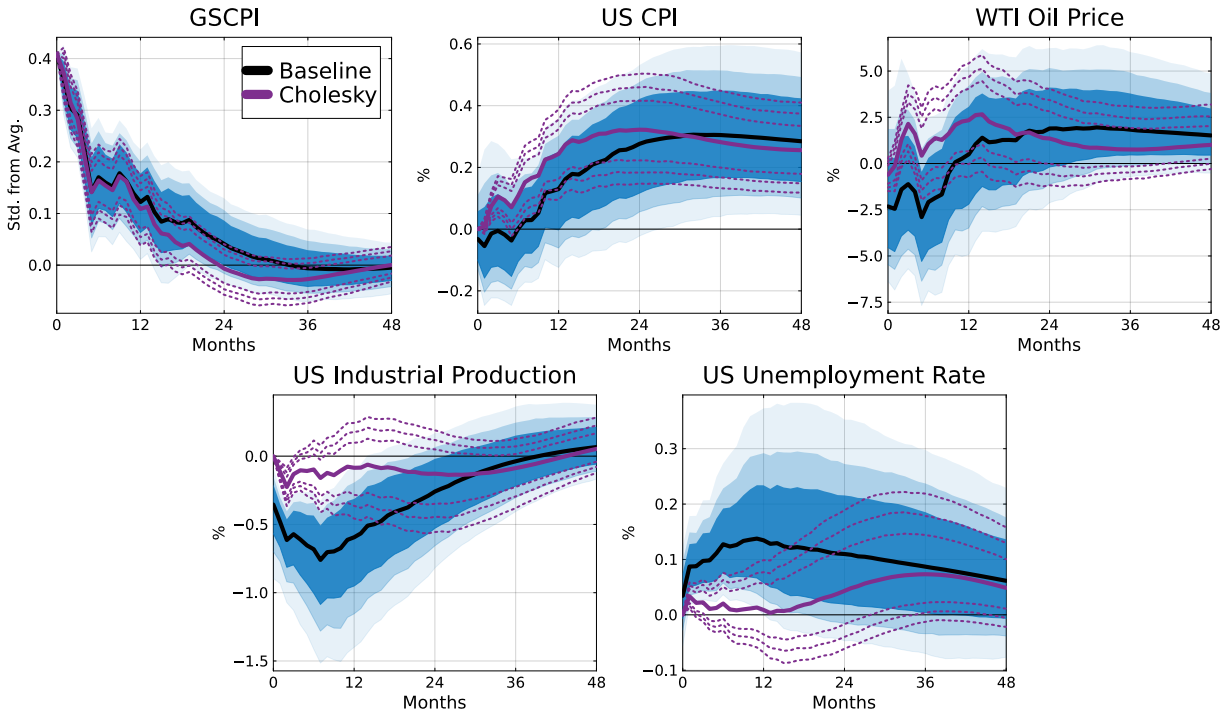


Figure C.9: Impulse Responses: Recursive vs External Instrument Identification

Note: Impulse responses to a global supply chain shock corresponding to a $1,066 price surcharge. The solid black lines are the baseline external-instrument point estimates, with shaded areas denoting the 68, 80 and 90 percent confidence bands; the solid purple lines are the point estimates under the recursive ordering identification, with dotted purple lines the corresponding confidence bands.

the global supply chain shock identified by the recursive ordering mixes the two, offsetting the responses of real activity. Without purging, the demand contamination attenuates the effects on the real economy, where they disappear altogether.

This attenuation has direct implications for the historical decomposition. Figure C.10 reports the cumulative contribution of the identified shock to consumer prices and industrial production from January 2020 to December 2024. The contributions to consumer prices are comparable across the two identifications, whereas those to industrial production differ sharply. Under the recursive ordering, global supply chain shocks exerted no meaningful downward pressure on economic activity, with a contribution that is not statistically different from zero over most of the period. The recursive identification would thus imply that

global supply chain disruptions played essentially no role in the pandemic-era production shortfall and recovery.

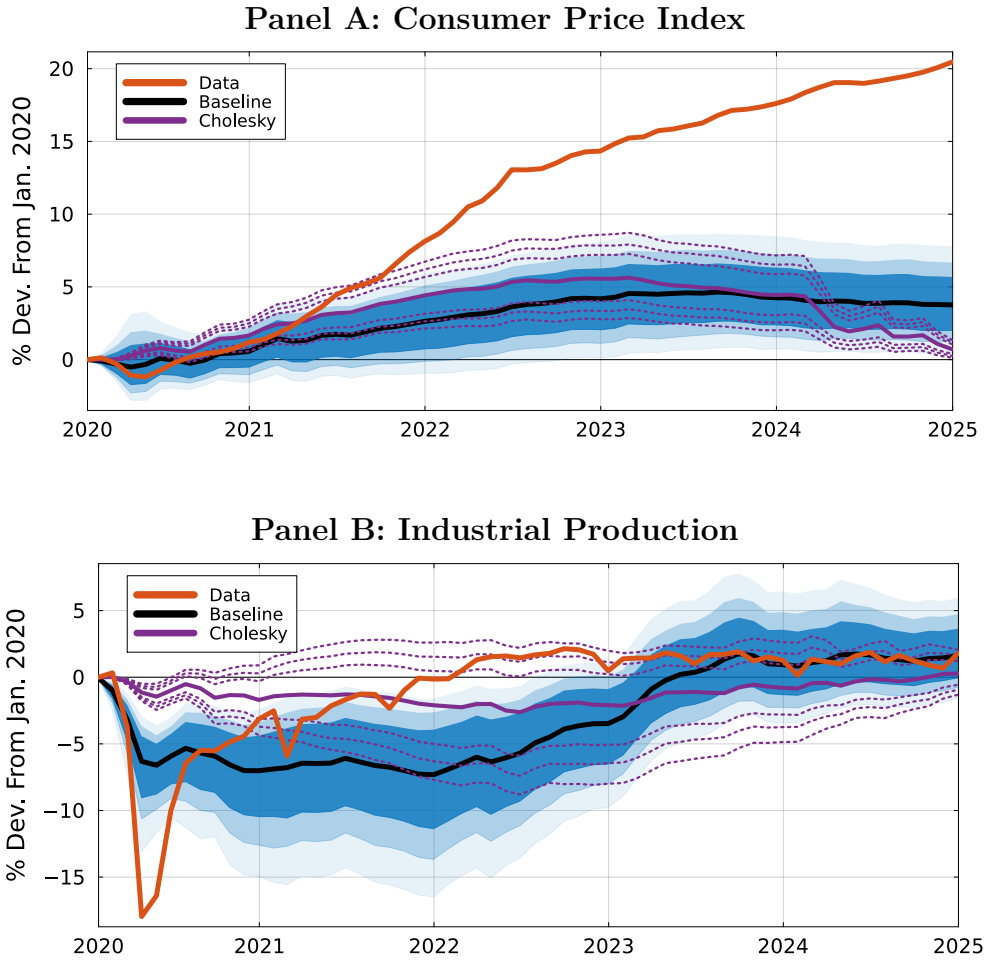


Figure C.10: Historical Decomposition: Recursive vs External Instrument Identification

Note: Historical contribution of global supply chain shocks to the US CPI (Panel A) and US Industrial Production (Panel B); both the actual series and the estimated contributions are expressed as percentage deviations from the pre-pandemic level (January 2020). The solid orange line is the actual series; the solid black line and shaded areas are the point estimate and 68, 80 and 90 percent confidence bands of the baseline Proxy-SVAR contribution; the solid purple line and dotted purple lines are the point estimate and confidence bands of the contribution under the recursive ordering identification.

To corroborate this interpretation, Table C.7 reports the forecast error variance decomposition under the two identification strategies. On the nominal side, the two identifications attribute a comparable share of consumer price variability to the shock. On the real side, the recursive shock accounts for a negligible share of the variance of industrial production

and unemployment. Under the recursive identification, global supply chain shocks are quantitatively important only in explaining the fluctuations of consumer prices.

Table C.7: Forecast Error Variance Decomposition: Recursive vs External Instrument

	US CPI		**WTI Oil Price**		**US Ind. Prod.**		**US Unemp.**		**GSCPI**	
	(RO)	(IV)	(RO)	(IV)	(RO)	(IV)	(RO)	(IV)	(RO)	(IV)
Horizon										
0	0.00	0.01	0.00	0.04	0.00	0.16	0.00	0.01	0.97	0.78
12	0.11	0.02	0.03	0.02	0.01	0.25	0.00	0.13	0.92	0.74
24	0.28	0.11	0.05	0.03	0.01	0.24	0.00	0.11	0.77	0.66
36	0.37	0.21	0.05	0.05	0.02	0.24	0.02	0.11	0.70	0.60
48	0.40	0.26	0.05	0.06	0.02	0.24	0.03	0.11	0.69	0.59
∞	0.39	0.30	0.06	0.08	0.05	0.24	0.04	0.10	0.69	0.58

Note: The table compares the proportion of the forecast error variance of the variables in the baseline model explained by the global supply chain shock under the recursive ordering identification (RO, left columns) and the external instrument identification (IV, right columns) at horizons 0, 12, 24, 36, 48 months and asymptotically.

Taken together, the differences between the two identifications are large and systematic, and endogeneity emerges as a first-order concern. Under the recursive ordering, the estimated real effects of global supply chain shocks are muted, the contribution to the pandemic-era production shortfall disappears, and the shocks no longer emerged as key drivers of business-cycle fluctuations as a whole.

D Data

This section provides details on the variables used in the aggregate and sectoral analysis, as well as those included in the robustness check involving additional control variables. Specifically, Table D.8 lists the series ID, series name, data source, and the transformation applied to each variable. The table is divided into panels, each containing the variables used in a specific analysis, as indicated by the corresponding panel title. All variables in Panels B through E were added individually to the baseline specification to test the robustness

of the main results to additional controls (Figure B.7). Variables used exclusively in the Granger causality tests (Table A.1) or the correlation analysis with other structural shocks (Table A.2) are not included, as their sources are already documented in the corresponding tables. All variables were used over the sample period from January 1998 to December 2024.

Table D.8: Data description

Series ID	Series Name	Data Source	Transformation
Panel A. Aggregate Analysis			
GSCPI@NY	Global Supply Chain Pressure Index	New York FED	Level
S111PC@G10	US CPI	Haver Analytics	100 log
N111POIL@G10	WTI Oil Price	Haver Analytics	100 log
S111D@G10	US Industrial Production	Haver Analytics	100 log
S111ELUR@G10	US Unemployment Rate	Haver Analytics	Level
Panel B. Additional Controls			
	Oil Supply News Shock	Känzig (2021)	Level
	Oil Supply Shock	Baumeister and Hamilton (2019)	Level
MTSO133FMS	Federal Outlays Growth	FRED	YoY Growth
H111PFTT@G10	US Terms of Trade	Haver Analytics	100 log
FFR	Fed Funds Rate	FRED	Level
CINF5@USECON	Inflation Expectations 5Y	Haver Analytics	Level
CINF1@USECON	Inflation Expectations 1Y	Haver Analytics	Level
S001DPP@G10	World Industrial Production	Haver Analytics	100 log
CUSR0000SEHF	US CPI Energy Services	FRED	100 log
PCUCC@USECON	US CPI Commodities	Haver Analytics	100 log
S111PCH@G10	US CPI Recreation	Haver Analytics	100 log
PCUHF@USECON	US CPI Fuels and Utilities	Haver Analytics	100 log
S111PCT@G10	US CPI Transportation	Haver Analytics	100 log
S111PCO@G10	US CPI Others	Haver Analytics	100 log
PCUSE@USECON	US CPI Energy	Haver Analytics	100 log
PCUTV@USECON	US CPI Motor Vehicles	Haver Analytics	100 log
PCUA@USECON	US CPI Apparel	Haver Analytics	100 log
UHFGE@USECON	US CPI Household Electricity	Haver Analytics	100 log
UHFGU@USECON	US CPI Piped Gas Service	Haver Analytics	100 log
PPIACO	US PPI	FRED	100 log
PCUOMFGOMFG	US PPI Manufacturing	FRED	100 log
PCU311311	US PPI Food Manufacturing	FRED	100 log

IP521@USECON	US IP Business Equipment	Haver Analytics	100 log
IPHT@USECON	US IP High Tech Industries	Haver Analytics	100 log
IP54@USECON	US IP Nonindustrial Supplies	Haver Analytics	100 log
IPUTL@USECON	US IP Electric and Gas Utilities	Haver Analytics	100 log
IPG3@USECON	US IP Machinery	Haver Analytics	100 log
IPG61@USECON	US IP Motor Vehicles	Haver Analytics	100 log
IPF1@USECON	US IP Wood	Haver Analytics	100 log
IPF7@USECON	US IP Nonmetallic Mineral Products	Haver Analytics	100 log
IPG1@USECON	US IP Primary Metals	Haver Analytics	100 log
IPG2@USECON	US IP Fabricated Metal Products	Haver Analytics	100 log
IPG4@USECON	US IP Comp and Elec Comp	Haver Analytics	100 log
IPG5@USECON	US IP Elec Eqpt, Appliances	Haver Analytics	100 log
IPG64T9@USECON	US IP Aerospace	Haver Analytics	100 log
IPG7@USECON	US IP Furniture	Haver Analytics	100 log
IPE1T2@USECON	US IP Food Bev and Tob	Haver Analytics	100 log
IPE3T4@USECON	US IP Textile	Haver Analytics	100 log
IPE5T6@USECON	US IP Apparel	Haver Analytics	100 log
IPF2@USECON	US IP Paper	Haver Analytics	100 log
IPF3@USECON	US IP Printing and Rel	Haver Analytics	100 log
IPF5@USECON	US IP Chemicals	Haver Analytics	100 log
IPF4@USECON	US IP Petroleum and Coal	Haver Analytics	100 log
IPF6@USECON	US IP Plastics and Rubber	Haver Analytics	100 log
IPN221113S	US IP Nuclear El Gen	FRED	100 log
IPB51222S	US IP Residential En Utilities	FRED	100 log
IPG2211S	US IP Electric Power	FRED	100 log
IPG2212S	US IP Gas Distribution	FRED	100 log
NMFVDI@USECON	Services Supplier Deliveries Index	Haver Analytics	100 log
NRSV2@USECON	Food Drinking Services	Haver Analytics	100 log
	Real Food Services (CPI Defl.)	Own Calculation	100 log
	Real Food Services (CPI Food Defl.)	Own Calculation	100 log
NAPMC@USECON	Manufacturing Composite Index	Haver Analytics	100 log
NAPMEI@USECON	Manufacturing Employment Index	Haver Analytics	100 log
Panel C. Sectoral Production Analysis			
IPMFG@USECON	US IP Manufacturing	Haver Analytics	100 log
IPB0@USECON	US IP Mining	Haver Analytics	100 log
IPMDG@USECON	US IP Durable Goods	Haver Analytics	100 log
IPMND@USECON	US IP Nondurable Goods	Haver Analytics	100 log
IPE1T2@USECON	US IP Food and Beverages	Haver Analytics	100 log

IPG61T3@USECON	US IP Motor Vehicles and Parts	Haver Analytics	100 log
IP53@USECON	US IP Materials	Haver Analytics	100 log
IPFP@USECON	US IP Final Products	Haver Analytics	100 log
Panel D. Non-Tradable Analysis			
NMFBAIA@USECON	Services Activity Index	Haver Analytics	100 log
NMFEIA@USECON	Services Employment Index	Haver Analytics	100 log
NAPMOI@USECON	Manufacturing Production Index	Haver Analytics	100 log
Panel E. Sectoral Prices Analysis			
CPILFESL	US CPI Core	FRED	100 log
PCUCS@USECON	US CPI Services	Haver Analytics	100 log
PCUCCD@USECON	US CPI Durable Goods	Haver Analytics	100 log
PCUSND@USECON	US CPI Nondurable Goods	Haver Analytics	100 log
S111PCFB@G10	US CPI Food and Beverages	Haver Analytics	100 log
PCUTPV@USECON	US CPI New Vehicles	Haver Analytics	100 log
S111PCH@G10	US CPI Housing	Haver Analytics	100 log
S111PCM@G10	US CPI Medical Care	Haver Analytics	100 log

E Price Surcharges & Narrative Analysis

This appendix describes the two-step narrative analysis used to isolate 54 price surcharges caused by exogenous, supply-side disruptions to the global supply chain. Before illustrating the algorithm, Section E.1 gives an overview of the composition of these price surcharges, showing that routine repricing is pervasive, accounts for a large share of price surcharges, and travels through its own dedicated channel, distinct from the exogenous supply side disruption surcharges isolated in the instrument. Step 1 then applies a keyword-based filtering algorithm in the spirit of Baker, Bloom and Davis (2016) to the universe of 7,872 surcharge announcements issued by the three largest container carriers between 2014 and 2024, retaining only those whose announcement language is consistent with an exogenous supply-side disruption. Step 2 then performs an event-by-event cross-validation of each surviving candi-

date against external sources, local press, industry reports, port authority statements, and academic references to confirm both the factual accuracy of the disruption and its exogenous, supply-side character. For every confirmed event, I report three pieces of evidence: (i) the technical details of the surcharge; (ii) the background of the affected port or trade route immediately before the event; and (iii) the proximate cause of the surcharge, emphasising its exogenous, unanticipated, supply-side nature, unrelated to other structural or demand-side shocks affecting the global economy at the time.

E.1 The Composition of Surcharge Announcements

Before applying the filtering algorithm, it is useful to document what the universe of announcements contains. The key point that I want to illustrate is that carriers do not need an exogenous disruption to change prices: as oligopolists they reprice to demand and supply conditions continuously, and they do so through dedicated price surcharges.

I adopt the most conservative classification available, and I keep only two separate groups. The first is the peak-season surcharge (PSS), levied when demand peaks against available capacity and therefore unambiguously a response to demand. The second, which I label general rate changes (GRC), collects freight-all-kinds (FAK) rates, general rate increases and restorations, and rate and price announcements; these respond to demand and to supply conditions jointly, and the announcement text does not allow the two to be separated. An announcement counts only when it explicitly names one of these charges: everything else is excluded, including surcharges that in principle may still embed a response to demand, such as congestion and canal-transit fees. Where a carrier heads an announcement with the trade lane alone, it counts only when the opening of the text names the charge.

Table E.10 reports the result. Even under this restrictive definition, 2,794 announcements, 35.5% of the total, are routine repricing: roughly one every business day over the sample, against the 54 exogenous disruptions the narrative analysis retains. Peak-season surcharges

Table E.10: Composition of price surcharges.

Carrier	Period	Total	PSS	GRC	Combined
MSC	Jan 2014–Dec 2024	1,217	12.6%	21.4%	33.9%
Maersk	Mar 2019–Dec 2024	2,713	14.3%	4.9%	19.2%
CMA CGM	Jan 2014–Dec 2024	3,942	11.5%	35.7%	47.2%
Total	Jan 2014–Dec 2024	7,872	12.7%	22.8%	35.5%

Note: The table reports, for each carrier, the period over which price surcharges are collected (*Period*), the number of surcharges collected (*Total*), and the share of those surcharges classified as peak-season surcharges (*PSS*), as general rate changes such as freight-all-kinds rates, general rate increases and restorations, and rate and price announcements (*GRC*). The two groups are mutually exclusive, so *Combined* is their sum.

alone account for 996 of them, 12.7% of all announcements; the remaining 1,798, 22.8% of the total, are general rate changes. The two groups are mutually exclusive: an announcement naming a peak-season surcharge is assigned to the first group even when it also names another charge.

The peak-season surcharge is by construction a response to demand, so the 12.7% is a floor on routine demand-driven repricing that does not rest on any interpretation of the carriers' language. The general rate changes cannot be attributed to demand alone, since a carrier raising a freight-all-kinds rate may be responding to firm demand, to tightening capacity, or to both at once. Nothing in the argument turns on separating them: what matters is that carriers reprice continuously through channels that require no exogenous disruption, whichever conditions drive them.

The figure of 35.5% is a lower bound. It is most accurate for CMA CGM, which names the surcharge type almost without exception; there the share is 47.2%, the highest of the three carriers. Maersk's lower share reflects its practice, before 2024, of titling rate announcements with the trade lane alone and naming the charge only deeper in the body, beyond the opening that the classification reads. The measurement error therefore runs in one direction, and the

true share of routine repricing is higher than the aggregate reported here.

E.2 Step 1: Isolate Exogenous Supply-Side Events from Price Announcements

Starting from the full set of 7,872 surcharge announcements published by the world's three largest container shipping companies between 2014 and 2024, I adapt the keyword-based filtering algorithm of Baker, Bloom and Davis (2016) to isolate surcharges that include terms commonly associated with exogenous supply-side disruptions. I specify keywords related to the four broad categories of disruption described earlier:

1. **Operational disruptions:** pcs, disruption, congestion, long-time container dwell, late gate, blockage, operation cost recovery, driver, safe working practice, sanitary measure, weight charge, demolition, additional surcharge, temporary increase, stop receiving, reroute, divert, fire, damage.
2. **War-related disruptions:** emergency, extraordinary, risk, war, conflict, attack, pirate.
3. **Labor strikes and industrial action:** strike, industrial action, force majeure, work condition.
4. **Natural disasters and weather-related disruptions:** low water, water level, earthquake, drought, water conservation, cyclone, heavy rain.

To avoid inflated match counts, I search for core word stems only, treating singular and plural forms, as well as inflected verb forms, as equivalent. This automated filtering yields a set of 369 candidate announcements. Each is then subject to narrative analysis and external validation to confirm whether it reflects a genuinely exogenous supply-side disruption. One reason for exclusion is that many instances of port congestion are driven by high demand, rather than by supply-side disruptions. In addition, a considerable number of announcements

are excluded because keywords appear in generic tariff breakdowns or standardized language used to describe pricing structures across routes, rather than to justify a surcharge linked to a specific event. Another reason for exclusion is the lack of supporting evidence: in some cases, announcements refer to congestion without providing any concrete cause, and no further explanation can be found in local newspapers, industry reports, or port authority statements. I also exclude one port congestion surcharge attributed to reduced labor availability in the UK, as worker representatives cited in local media explicitly describe the surcharge as profit-driven and not reflective of actual port conditions.

Table E.11 presents all 54 exogenous supply-side disruptions retained after the second step. For each event, I report the announcement and implementation dates, the source, the affected ports or trade routes, and the corresponding surcharge. To ensure that the keyword-based algorithm is not systematically missing relevant events, I manually review each of the 7,872 surcharge announcements. This comprehensive review reveals three disruptions that are not captured by the algorithm but reflect clear exogenous supply-side shocks. First, I identify an announcement related to a new low-sulfur regulation in Chinese ports, which contains no keywords from the filtering list. However, supporting evidence from Tan et al. (2021) and Weng et al. (2022) documents how the unanticipated and unexpected expansion of the Emission Control Areas (ECAs), with the subsequent introduction of a low-sulfur regulation in China, incentivizes evasive routing behavior by shipping companies, thereby increasing shipping times. Second, I find two distinct events related to widespread inland transportation congestion across India's dry port network.

Table E.11: Collection of all the exogenous events

Key Dates	Ports Involved	Source	Surcharge	Keywords
Port Congestion After Shipping Relocation in Sohar, Oman				
Announcement: 2014-08-25 Implementation: 2014-08-25	Europe to Sohar, Oman	MSC	**200 USD**	Extraordinary = 1, work condition = 1
War and Naval Blockade in Hodeidah, Yemen				
Announcement: 2015-05-19 Implementation: 2015-05-19	All Trade to Hodeidah	CMA CGM	**600 USD**	risk = 5, war = 2

Crane Operators Strike at Nhava Sheva				
Announcement: 2015-08-07 Implementation: 2015-08-15	From all Asian Ports to/via Nhava Sheva	CMA CGM	**300 USD**	emergency = 5, force majeure = 1
Crane Failure and Sand Storm Disrupt Shuwaik, Kuwait				
Announcement: 2015-09-01 Implementation: 2015-09-01	From/to Shuwaikh	CMA CGM	**200 USD**	disruption = 2, congestion = 6
Airstrikes and Port Closure, Hodeidah				
Announcement: 2015-09-01 Implementation: 2015-09-01	To all cargo initially planned for Hodeidah (Bab-el-Mandeb)	CMA CGM	**400 USD**	stopped receiving = 1
Airstrikes and Port Closure, Hodeidah				
Announcement: 2015-10-21 Implementation: 2015-10-21	On Asia - Europe Mediterranean Trade	CMA CGM	**500 USD**	risk = 5, war = 2
Road Closure and Cargo Diversion, Kuwait				
Announcement: 2016-08-26 Implementation: 2016-08-26	World to Shuwaikh and Shuaiba, Kuwait	MSC	**600 USD**	congestion = 2, demolition = 1
Road Closure and Cargo Diversion, Kuwait				
Announcement: 2016-09-06 Implementation: 2016-10-01	From/to Shuwaikh	CMA CGM	**600 USD**	disruption = 2, congestion = 6, demolition = 1
Nationwide Port Strike Disrupts Cargo in Spain				
Announcement: 2017-03-07 Implementation: 2017-03-10	All Spanish ports on Europe US Trade	CMA CGM	**200 USD**	pcs = 1, congestion = 2, emergency = 2
Road Closure and Cargo Diversion, Kuwait				
Announcement: 2017-08-21 Implementation: 2017-08-20	From/to Shuwaikh	CMA CGM	**200 USD**	congestion = 6
Battle of Aden and Port Closure, Yemen				
Announcement: 2018-01-09 Implementation: 2018-01-08	From/to Aden	CMA CGM	**600 USD**	pcs = 1, congestion = 6
Aden, Yemen Shipping Improves After UN Aid Plan				
Announcement: 2018-04-13 Implementation: 2018-04-13	From/to Aden	CMA CGM	**-200 USD**	pcs = 1, congestion = 5
Tropical Storm and Monsoon Flooding Disrupt Yangon, Myanmar				
Announcement: 2018-06-01 Implementation: 2018-06-01	Imports into all Myanmar ports	CMA CGM	**200 USD**	congestion = 6
Low Water Surcharge St. Lawrence Seaway, Canada				

Announcement: 2018-07-16 Implementation: 2018-08-01	From Europe and Mediterranean to Canada East Cost	CMA CGM	**150 USD**	low water = 2
Port Worker Strikes Disrupt Shipping in Portugal				
Announcement: 2018-09-18 Implementation: 2018-10-18	Export from Lisbon	CMA CGM	**150 USD**	disruption = 2, congestion = 3, emergency = 1, pcs = 1
Shuwaikh Congestion from Dust Storms				
Announcement: 2018-09-25 Implementation: 2018-10-25	From/to Shuwaikh	CMA CGM	**300 USD**	disruption = 2, congestion = 6
Aden Port Congestion from Yemen Conflict				
Announcement: 2018-11-28 Implementation: 2018-12-01	From/to Aden	CMA CGM	**400 USD**	pcs = 1, congestion = 5
Regulatory Shock Across China, Hong Kong, and Taiwan				
Announcement: 2018-12-14 Implementation: 2019-01-01	Ex/via China, Hong Kong, Taiwan to all destinations worldwide	CMA CGM MSC	**41.25 USD**	missing
Low Water Level St. Lawrence Seaway, Canada				
Announcement: 2018-12-20 Implementation: 2019-01-17	From Europe and Mediterranean to Canada East Cost	CMA CGM	**125 USD**	low water = 2
Gulf Shipping Disruptions from Strait of Hormuz Attack				
Announcement: 2019-07-02 Implementation: 2019-08-01	From/to Bahrein, Iraq, Kuwait, Oman, Qatar, Saudi Arabia, UAE	CMA CGM MSC	**76 USD**	risk = 2 (CMA CGM); risk = 4, war = 4 (MSC)
UAE Military Drawdown Triggers Congestion at Aden				
Announcement: 2019-07-19 Implementation: 2019-08-20	From/to Aden	CMA CGM	**300 USD**	operation cost recovery = 5
Transport Strikes Disrupt Shipping in South France				
Announcement: 2020-01-22 Implementation: 2020-02-01	From/to Southern France	Maersk	**425 USD**	disruption = 4, congestion = 2, emergency = 1, risk = 1, strike = 2
Port Congestion from Secure Workplace Protocols, Felixstowe				
Announcement: 2020-09-22 Implementation: 2020-10-01	From/to Felixstowe to/from Far East and South East Asia	CMA CGM	**300 USD**	congestion = 5, driver = 1
Australia Port Congestions from Strikes				

Announcement: 2020-09-08 Implementation: 2020-10-08	From/to Sidney, Melbourne, Brisbane	CMA CGM Maersk MSC	**556.7 USD**	pcs = 1, congestion = 3, emergency = 2, industrial action = 1 (CMA CGM); congestion = 5, industrial action = 1 (Maersk); congestion = 5, industrial action = 1, disruption = 1 (MSC)
Bilbao Port Strike Escalation				
Announcement: 2020-11-07 Implementation: 2020-11-16	From/to Bilbao	CMA CGM MSC	**250 USD**	pcs = 1, congestion = 5 (CMA CGM); congestion = 5, strike = 1 (MSC)
Automated Straddle Failures Disrupt Auckland				
Announcement: 2020-11-03 Implementation: 2020-12-01	From/to Auckland	CMA CGM, MSC, MARSK	**476 USD**	congestion = 23, disruption = 1, pcs = 1
Lockdown Induced Congestions Across UK Container Gateways				
Announcement: 2020-11-19 Implementation: 2020-12-01	From/to all UK ports to/from Asia	CMA CGM	**240 USD**	pcs = 1, congestion = 6
Empty-Return Refusal and Chassis Gridlock at LA–LB Ports				
Announcement: 2020-11-04 Implementation: 2020-12-04	Deliveries made by truck or rail to Chicago, Houston, Columbus, Cincinnati, Dallas, Memphis, Nashville, St Louise, Minneapolis, LA-LB, Seattle, Oakland, Tacoma	MSC	**350 USD**	congestion = 1, emergency = 4
Low Water Levels St. Lawrence Seaway, Canada				
Announcement: 2021-01-11 Implementation: 2021-02-07	From /to Canada East Cost	CMA CGM	**50 USD**	low water = 2, water level = 1
UK Surcharge on Indian Imports Due to Colombo Transshipment Breakdown				
Announcement: 2021-03-12 Implementation: 2021-03-15	From India to Southampton and London Gateway	CMA CGM	**400 USD**	pcs = 1, congestion = 6
Suez Canal Blockage Surcharge and Extensions				
Announcement: 2021-05-07 Implementation: 2021-05-07	From/to Europe and Mediterranean	Maersk	**42.5215 USD**	congestion = 1, blockage = 1, extraordinary = 1
Suez Canal Blockage Surcharge and Extensions				

Announcement: 2021-05-07 Implementation: 2021-06-01	From/to Europe and Mediterranean	Maersk	**42.1435 USD**	congestion = 1, blockage = 1, extraordinary = 1
Low Water Surcharge, St. Lawrence Seaway, Canada				
Announcement: 2021-06-01 Implementation: 2021-06-15	From Europe and Mediterranean to Canada	CMA CGM Maersk MSC	**266 USD**	low water = 2, water level = 1 (CMA CGM); low water = 2, water level = 2 (Maersk); low water = 3, water level = 1 (MSC)
Suez Canal Blockage Surcharge and Extensions				
Announcement: 2021-06-21 Implementation: 2021-07-01	From/to Europe and Mediterranean	Maersk	**41.3875 USD**	extraordinary = 1
Suez Canal Blockage Surcharge and Extensions				
Announcement: 2021-07-21 Implementation: 2021-08-01	From/to Europe and Mediterranean	Maersk	**41.118 USD**	extraordinary = 1
Meishan Terminal Closure, Ningbo				
Announcement: 2021-08-03 Implementation: 2021-08-15	All shipment from Asia to Europe and Mediterranean	MSC	**1000 USD**	disruption = 4
Ocean Carriers Refusal to Accept Empty Container Returns Generate Disruptions, LA-LB				
Announcement: 2021-11-03 Implementation: 2021-11-01	Ports of Los Angeles and Long Beach	MSC	**100 USD**	disruption = 2, congestion = 2, emergency = 1
Ocean Carriers Refusal to Accept Empty Container Returns Generate Disruptions, LA-LB				
Announcement: 2021-12-01 Implementation: 2021-12-29	Ports of Los Angeles and Long Beach	Maersk	**100 USD**	temporary increase = 1
War Disrupt Black Sea Trade				
Announcement: 2022-03-02 Implementation: 2022-04-03	From/to all Black Sea Ports	Maersk	**1100 USD**	emergency = 8, risk = 8, conflict = 1
Equipment Delays and Strike Hit Le Havre				
Announcement: 2022-04-20 Implementation: 2022-04-19	Port of Le Havre TDF	Maersk	**135.4375 USD**	congestion = 3, extraordinary = 1
India Dry Port Chaos After Price Control Regulation				
Announcement: 2022-05-02 Implementation: 2022-06-01	All Indian locations via Indian Ports to World	Maersk	**24.71 USD**	missing
Strikes in Northern Europe				
Announcement: 2022-09-07 Implementation: 2022-09-07	Ports of Le Havre TDF, Dunkirque	Maersk	**122.5 USD**	congestion = 2, extraordinary = 1

Diversion Over Strike Fears Overwhelms Port Houston

Announcement: 2022-11-15 Implementation: 2022-12-01	Port of Houston	Maersk	**45 USD**	late gate = 9

ILWU Work Refusal on Pier 400 Automation, LA–LB

Announcement: 2022-11-22 Implementation: 2022-12-01	Ports of Los Angeles and Long Beach	Maersk	**270 USD**	late gate = 9

India Dedicated Freight Corridor Expansion

Announcement: 2023-01-23 Implementation: 2023-02-20	All Indian locations via Indian Ports to World	Maersk	**-250.85 USD**	missing

Disruptions at Turkish Ports Following Earthquake

Announcement: 2023-03-27 Implementation: 2023-03-27	From Mersin to all destinations worldwide	CMA CGM	**100 USD**	pcs = 3, congestion = 2

Disruptions at Turkish Ports Following Earthquake

Announcement: 2023-04-07 Implementation: 2023-04-07	From North East Asia, South East Asia and Hong Kong SAR to Mersin	CMA CGM	**100 USD**	pcs = 2, congestion = 1

Panama Canal Surcharge Due to Drought

Announcement: 2023-04-06 Implementation: 2023-06-01	From Northeast Asia, China to East Coast Central America	CMA CGM	**600 USD**	low water = 2, water level = 1

Panama Canal Congestion Due to El Niño Drought

Announcement: 2023-07-07 Implementation: 2023-08-03	All cargo transiting the Panama Canal	Maersk	**240 USD**	emergency = 1, risk = 1

Disruptions at Turkish Ports Following Earthquake

Announcement: 2023-08-25 Implementation: 2023-09-01	From/to Iskenderun Turkey to/from Europe and Middle East	CMA CGM	**207.6 USD**	pcs = 2, congestion = 2

Further Restrictions Panama Canal Due to El Niño Drought

Announcement: 2023-11-23 Implementation: 2023-12-15	All cargo transiting the Panama Canal	CMA CGM MSC	**298.5 USD**	drought = 1, precipitation = 1, water conservation = 1 (CMA CGM); emergency = 2, water level = 1 (MSC);

Red Sea Conflict Forces Cape of Good Hope Rerouting

Announcement: 2023-12-20 Implementation: 2024-01-01	From/to Europe and Mediterranean to/from Asia	MSC	**891.67 USD**	re-routing = 2 (CMA CGM); disruption = 8, emergency = 4, risk = 1, re-routing = 1, divert = 5, diverting = 1 (Maersk); divert = 1 (MSC)
Low Water Levels St. Lawrence Seaway, Canada				
Announcement: 2024-10-21 Implementation: 2024-11-01	From Europe and Mediterranean to Montreal, Canada	CMA CGM Maersk MSC	**400 USD**	low water = 4, water level = 1 (MSC); pcs = 5, congestion = 5 (CMA CGM); low water = 4 (Maersk)
Low Water Levels St. Lawrence Seaway, Canada				
Announcement: 2024-11-04 Implementation: 2024-12-01	All cargo transiting the Panama Canal	CMA CGM Maersk	**300 USD**	low water = 1 (CMA CGM); emergency = 2, risk = 2, low water = 4 (Maersk)

E.3 Step 2: Corroborate Price Surcharges with External Sources

For each event below I follow a uniform three-paragraph template. The first paragraph, *Technical details of the surcharge*, reports the announcement and implementation dates, the issuing carrier, the affected ports or trade routes, the surcharge rate, and the broad disruption category (operational, war-related, strike, or weather-related). The second paragraph, *Background*, describes the institutional and structural conditions at the affected port or along the affected route in the period leading up to the event. The third paragraph, *Cause of the price surcharge*, identifies the disruption and explains why it constitutes an exogenous, unanticipated, supply-side shock to shipping capacity, unrelated to contemporaneous demand-side dynamics or other structural macroeconomic shocks.

Port Congestion After Shipping Relocation in Sohar, Oman (August 2014)

Technical details of the surcharge.— Announced and implemented on 25 August 2014 by MSC, the surcharge applied to cargo on the Europe-to-Sohar (Oman) route at a rate of $200 USD per FEU (MSC Notice). The event is classified as an operational disruption.

Background.— In 2014, the Omani government decided to convert Port Sultan Qaboos in Muscat, the country's main commercial cargo port, into a cruise terminal to relieve road congestion in the capital (Port Sultan Qaboos, Wikipedia). All commercial shipping was to be redirected to the Port of Sohar, 230 km northwest of Muscat, by a hard government deadline of 31 August 2014 (CMA CGM Notice, Logistics Middle East). The receiving terminal at Sohar had just completed a major expansion in May 2014 (Dredging Today), but had been operational for fewer than three months and was still ramping up equipment and workflows at scale (Marine Link).

Cause of the price surcharge.— Immediately before the 31 August deadline, Eid al-Fitr fell on 28–29 July 2014 (Calendar Date), a major public holiday across the Gulf that caused widespread staffing gaps, delayed return of truck drivers and dock workers, and misaligned schedule resumptions across customs offices and terminal operators, with no logistics actor able to plan around it (Logistics Middle East). It was precisely into this labor vacuum that the first substantial cargo volumes arrived at an immature terminal. The resulting operational breakdown was not an inference: Oman International Container Terminal's own CEO acknowledged a *"premature peak in cargo volumes"* driven by a faster-than-anticipated shift of operations to Sohar (Journal of Commerce), confirming that the terminal had been overwhelmed before it had the institutional capacity to absorb the load. A simultaneous accumulation of empty containers further consumed yard space and suppressed throughput. It was precisely this August congestion, coinciding with the final days of the transition, that led MSC to issue its Port Congestion Surcharge on 25 August 2014 (MSC Customer Advisory), a shock to shipping capacity entirely unrelated to any increase in import demand

from Oman.

War and Naval Blockade in Hodeidah, Yemen (May 2015)

Technical details of the surcharge.— Announced and implemented on 19 May 2015 by CMA CGM, the surcharge applied to all trade to Hodeidah (Yemen) at a rate of $600 USD per FEU (CMA CGM Notice). The event is classified as a war-related disruption.

Background.— Hodeidah is Yemen's principal commercial port, handling approximately 70–80% of food and humanitarian supplies for northern Yemen (Washington Institute for Near East Policy), with no realistic substitute port for the north (Hudaydah Port, Wikipedia). Yemen had been experiencing an internal conflict between the Houthi movement and government forces since late 2014 (Saudi-led Intervention in Yemen, Wikipedia), a situation shipping markets had broadly been monitoring. The Bab al-Mandab strait, through which all Hodeidah-bound vessels must pass, is a critical and narrow chokepoint connecting the Red Sea to the Gulf of Aden.

Cause of the price surcharge.— What transformed the monitored conflict into an acute supply-side shock was the sudden military intervention launched without any prior warning: on 26 March 2015, a Saudi-led coalition began airstrikes and imposed a naval blockade, sealing the Bab al-Mandab strait and physically cutting off all commercial access to Hodeidah (Blockade of Yemen, Wikipedia). The blockade was sustained through April and into May, with ships denied berthing for weeks despite repeated contact attempts: by 1 May 2015, seven vessels carrying fuel were anchored outside Yemeni waters awaiting permission to dock (Human Rights Watch). In April, risk in the Gulf intensified further with the unprecedented seizure of the Maersk Tigris by Iranian forces in the Strait of Hormuz (MV Maersk Tigris, Wikipedia), a vessel detained over a decade-old civil dispute entirely unrelated to Yemen trade (USNI News). It was the accumulation of these supply-side disruptions, sustained and compounding through April and into May, that led CMA CGM to formalize the War Risk Surcharge in May 2015.

Crane Operators Strike at Nhava Sheva (August 2015)

Technical details of the surcharge.— Announced on 7 August 2015 and implemented on 15 August 2015 by CMA CGM, the surcharge applied to cargo from all Asian ports to or via Nhava Sheva (India) at a rate of $300 USD per FEU (CMA CGM Notice). The event is classified as a strike-related disruption.

Background.— Jawaharlal Nehru Port (JNPT) at Nhava Sheva, near Mumbai, is India's largest container port (Jawaharlal Nehru Port, Wikipedia). Gateway Terminals India (GTI), a joint venture between APM Terminals and the Container Corporation of India, operates one of the port's main container terminals, equipped with 10 post-Panamax quay cranes and 40 rubber-tyred gantry cranes (The Dollar Business). APM Terminals had been relying on subcontractors rather than direct employees for operations and maintenance roles at GTI, a practice that had generated growing tensions with the local unions (Offshore Energy).

Cause of the price surcharge.— In late July 2015, RTG crane operators at GTI launched a "go slow" action that rapidly escalated into a full work stoppage, bringing terminal operations to a total standstill by early August (OOCL Customer Advisory). The strike was triggered by the longstanding dispute over outsourced employment: workers demanded that APM Terminals issue a fresh tender for the subcontracting arrangement and create pathways for permanent employment. The action was abrupt and caught carriers off guard, as the union's president later described how *"the terminal doors were closed to workers, operations stopped and vessels diverted away from Mumbai"* (Offshore Energy). With no berth availability at GTI, vessels were forced to omit Nhava Sheva entirely, with cargo redirected to Colombo (OOCL Customer Advisory). CMA CGM issued the Emergency Contingency Surcharge on 7 August 2015, a shock to shipping capacity driven by a labor dispute over employment terms, entirely unrelated to any change in import demand at Nhava Sheva.

Crane Failure and Sand Storm Disrupt Shuwaikh, Kuwait (September 2015)

Technical details of the surcharge.— Announced and implemented on 1 September 2015 by CMA CGM, the surcharge applied to cargo from and to Shuwaikh (Kuwait) at a rate of $200[1] USD per FEU (CMA CGM Notice). The event is classified as an operational disruption.

Background.— Shuwaikh Port is Kuwait's principal commercial seaport, situated on Kuwait Bay in Kuwait City and handling the bulk of the country's containerized imports (Shuwaikh Port, Wikipedia). The port faces chronic structural constraints: traffic congestion on internal and surrounding roads delays truck movements, a traffic law prohibiting trucks during rush hours forces vehicles to queue on inner roads, and container storage capacity is insufficient for sustained throughput (AlRukaibi, AlKheder and AlMashan (2020), Logistics Cluster). Kuwait is also one of the most dust-storm-prone countries in the Gulf, with severe events capable of reducing visibility to zero and forcing the Kuwait Ports Authority to suspend all maritime traffic at Shuwaikh, Shuaiba, and Doha ports simultaneously (Emirates 24/7).

Cause of the price surcharge.— In late August 2015, two of Shuwaikh's gantry cranes suffered unexpected mechanical failures when their cable reels were damaged, abruptly taking the units out of service and causing berthing delays of approximately five days per vessel (Container Management, 15 September 2015). The failures were unanticipated: the cranes had been in routine operation for years, and the simultaneous breakdown of two units caught terminal operators off guard, sharply reducing the port's physical cargo-handling capacity at a time when yard congestion from the customs bottleneck was already severe. In early September, a major dust storm swept across the Gulf region, with visibility in Kuwait dropping below half a nautical mile and winds exceeding 15 knots, forcing the Kuwait Ports Authority to halt all maritime traffic at Shuwaikh, Shuaiba, and Doha ports (Emirates 24/7).

[1]The price surcharge was increased from $200 USD per FEU, as announced in August 2015 by CMA CGM, to $400 USD per FEU. However, I was not able to find any sources about the cause for August's price surcharge, so I ignored it.

It was the combination of these two exogenous shocks, the crane failures in August and the dust storm in early September that led CMA CGM to issue the Port Congestion Surcharge on 1 September 2015, a disruption to shipping capacity entirely unrelated to any shift in Kuwait's import demand.

Airstrikes and Port Closure, Hodeidah (September - October 2015)

Technical details of the surcharges.— CMA CGM issued two successive surcharges on Yemen-related trade in the weeks following the destruction of Hodeidah port. Both events are classified as war-related disruptions.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
1 September 2015	1 September 2015	$400	CMA CGM Notice
21 October 2015	21 October 2015	$500[2]	CMA CGM Notice

Background.— Hodeidah had been under severe operational strain since the Saudi-led coalition launched its military intervention in Yemen in March 2015, imposing a naval blockade and prompting all major carriers to suspend direct port calls (Insurance Journal). Despite the blockade, reduced volumes of cargo continued to reach the port, which remained the only viable gateway for northern Yemen and handled the vast majority of food and humanitarian supplies for a population dependent on imports for most of its food consumption; food imports through Hodeidah in 2015 fell to roughly a third of the previous year's level and container throughput dropped by more than half (Washington Institute; Voice of America). By October 2015, the naval blockade had reduced commercial fuel imports to just one per cent of monthly requirements, and the Bab al-Mandab strait, through which all Asia–Europe Mediterranean shipping must transit, had become an active conflict zone.

Cause of the surcharges.— On 18 August 2015, the Saudi-led coalition carried out airstrikes directly targeting Hodeidah port, destroying all four main gantry cranes, the port control

[2]The price surcharge was increased from $400 to $900

room, walkways, grain silos, the customs building, and a World Food Programme warehouse, instantly rendering the port physically unable to receive or discharge cargo (ARWA Rights; Oxfam). With all cargo-handling equipment destroyed, CMA CGM redirected all Hodeidah-bound cargo to Aden, imposing substantial additional costs and causing delays, and issued the 1 September Operational Cost Recovery surcharge to recover those costs. In early October 2015, the conflict escalated further when Houthi forces launched anti-ship missile strikes against coalition naval vessels transiting the Bab al-Mandab: on 7 October a UAE Navy Baynunah-class corvette was struck, and on 10 October the Saudi replenishment ship *Yunbou* was targeted, both attacks using C-802 cruise missiles capable of reaching across the full width of the Red Sea (Washington Institute). These strikes demonstrated for the first time that commercial shipping transiting the strait faced a direct guided-missile threat, causing marine war risk insurance premiums to rise sharply and prompting CMA CGM to issue the 21 October War Risk Surcharge on all Asia–Europe Mediterranean cargo transiting the Bab al-Mandab. Both surcharges were shocks driven by the physical destruction of port infrastructure and the sudden militarisation of a critical maritime chokepoint, entirely unrelated to any change in Yemen's import demand.

Road Closure and Cargo Diversion, Kuwait (August 2016 – August 2017)

Technical details of the surcharges.— Multiple surcharges were issued on Kuwait-bound cargo following the closure of an important road infrastructure. All three events are classified as operational disruptions.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
26 August 2016	26 August 2016	\$600	MSC
6 September 2016	1 October 2016	\$600	CMA CGM
21 August 2017	20 August 2017	\$200[3]	CMA CGM

Background.— Shuwaikh Port is Kuwait's principal commercial port. The port's location in

[3]These values represent the overall price increase between the two ports of Shuwaikh and Shuaiba.

a densely populated urban area creates chronic road-access constraints: traffic congestion on internal and surrounding roads delays the movement of loaded trucks and the return of empty containers, while a traffic law prohibiting trucks during rush hours forces vehicles to queue on inner roads (AlRukaibi, AlKheder and AlMashan (2020)). The primary road link between Shuwaikh Port and the rest of Kuwait's highway network is the Ghazali Expressway, which connects the port area to Kuwait International Airport and the southern suburbs. Since 2010, the Jahra Road Development Plan, a major infrastructure project to convert the at-grade Jahra Road through the Shuwaikh industrial area into a two-level elevated motorway, had been under construction in phases, progressively restricting road capacity in the port's immediate vicinity. Shuaiba Port, located 45 kilometres south of Kuwait City, is Kuwait's second commercial port; its container terminal operates only four dedicated berths with a total quay length of 880 metres, handles roughly 110,000 TEU per year, and operates on a direct-delivery basis with no on-site storage facilities, leaving little buffer to absorb unexpected surges in vessel arrivals (ISA Kuwait).

Cause of the surcharges.— Although the Jahra Road Development Plan had been underway for years, the timeline for the most disruptive phase, the demolition of the Ghazali Bridge, was repeatedly postponed: local press had reported the closure as imminent as early as mid-2015, yet the bridge remained open for over a year beyond that date.[4] When the ramp connecting Jahra Road to the Ghazali Bridge was finally closed without prior warning in August 2016, the main road artery serving Shuwaikh Port was abruptly severed, catching carriers and trucking operators off guard (MSC; CMA CGM). The sudden loss of the primary access route forced all port-bound traffic onto secondary roads already congested by the ongoing construction, producing severe delays that sharply raised operational costs;

[4]The Ministry of Public Works held several meetings with the Chamber of Commerce, maritime unions, and port operators to discuss redirecting container ships from Shuwaikh Port to Shuaiba Port in response to the congestion near the port entrance. Al-Rai, 7 July 2015, original Arabic: *tafkīk jasr al-Ghazālī li-inshā' jisr ya'lūhu… lan yakūn qabl Aghusṭus* ("the dismantling of the Ghazali Bridge to build a bridge above it… will not be before August"); *taḥwīl ba'ḍ bawākhir al-ḥāwīyāt min mīnā' al-Shuwaykh ilā mīnā' al-Shu'ayba* ("redirecting some container ships from Shuwaikh Port to Shuaiba Port").

MSC issued the Port Congestion Surcharge on 26 August 2016, recommending that shippers redirect cargo to the alternative terminal at Shuaiba, while CMA CGM issued its own surcharge on 6 September 2016, effective 1 October 2016. The initial shock then propagated to Shuaiba: the Kuwait Shipping Companies and Agents Association coordinated the rerouting of at least eight container vessels from Shuwaikh to Shuaiba, with carriers including Maersk, CMA CGM, and APL redirecting services to the smaller port (Offshore Energy), and by mid-2017 the sustained influx of redirected vessels had overwhelmed Shuaiba's limited handling capacity: with only four container berths absorbing traffic previously spread across the larger Shuwaikh terminal, and no storage buffer to absorb the additional volume, berthing queues lengthened and yard backlogs mounted, leading CMA CGM to issue a further surcharge on 21 August 2017 covering both ports. Both surcharges were shocks driven by the physical severance of Shuwaikh's primary road access link and the consequent saturation of Shuaiba's fixed handling capacity, entirely unrelated to any change in Kuwait's import demand.

Nationwide Port Strike Disrupts Cargo in Spain (March 2017)

Technical details of the surcharge.— Announced on 7 March 2017 and implemented on 10 March 2017 by CMA CGM, the surcharge applied to all Spanish ports on the Europe–US trade at a rate of $200 USD per FEU (CMA CGM Notice). The event is classified as a strike-related disruption.

Background.— In 2014, the European Court of Justice ruled that Spain's port labor regime violated European Union freedom-of-establishment regulations by requiring terminal operators to hire dockers exclusively through local stevedore societies that held a de facto monopoly on port labor (The Maritime Executive). By mid-2016, Spain faced a daily penalty for noncompliance, prompting the government to introduce Royal Decree-Law 4/2017, which would gradually liberalize the hiring of port workers over four years (Port Today). The national stevedores' union rejected the reform, arguing it would eliminate thousands of dockworker

positions across Spanish ports (gCaptain, The Maritime Executive).

Cause of the price surcharge.— On 28 February 2017, the national stevedores' union announced nine days of strikes on alternating days between 6 and 24 March, the largest port strike in Spain since the 1980s, affecting all of the country's commercial ports simultaneously (Port Today). The stoppages were structured to halt cargo-handling operations every other hour beginning at 8 AM, with the port of Bilbao facing full 24-hour shutdowns on each strike day (gCaptain). CMA CGM's notice of 9 March 2017 stated that "shipping operations at Spanish ports have been badly hit due to labour disputes as a consequence of the Spanish Port Act" and that "industrial actions by Stevedores' Union with a number of strikes planned through the month of March will further deteriorate our operations in Spain". CMA CGM issued the Emergency Port Congestion Surcharge on 7 March 2017.

Battle of Aden and Port Closure, Yemen (January 2018)

Technical details of the surcharge.— Announced on 9 January 2018 and implemented on 8 January 2018 by CMA CGM, the surcharge applied to cargo from all ports worldwide to Aden (Yemen) at a rate of $600 USD per container (CMA CGM Notice). The event is classified as a war-related disruption.

Background.— Since the Saudi-led coalition's intervention in March 2015, Yemen's two main Red Sea ports, Hodeidah and Saleef, had faced escalating restrictions. Before the war, the two ports together handled roughly eighty per cent of all Yemeni imports (Human Rights Watch). The destruction of Hodeidah's gantry cranes in August 2015 severely curtailed the port's throughput, and on 6 November 2017 the coalition imposed a full blockade, shutting Hodeidah and Saleef entirely (Human Rights Watch). Although partially eased in subsequent weeks, the Red Sea ports remained severely restricted, forcing the bulk of commercial and humanitarian cargo to be rerouted to Aden (Human Rights Watch). By late 2017, the sustained diversion had produced severe berth congestion and extended vessel waiting times at Aden.

Cause of the price surcharge.— In January 2018, armed clashes erupted across Aden between government forces loyal to President Hadi and fighters of the Southern Transitional Council, rapidly escalating into a full-scale battle involving tanks and artillery in the districts of Khormaksar, al-Mansoura, and Dar Sad (Battle of Aden, Wikipedia). The fighting killed at least 38 people in three days and briefly forced the closure of Aden's seaport and airport (Human Rights Watch). The battle struck a port that was already operating far beyond its design capacity due to the blockade-driven cargo diversion, compounding existing delays and disrupting the schedules of vessels that had no alternative destination. CMA CGM issued the Port Congestion Surcharge on 9 January 2018, due to the combined effect of the blockade overflow and the armed conflict at Yemen's last functioning commercial port.

UN-Brokered Aid Plan Stabilises Aden Port Operations (April 2018)

Technical details of the surcharge.— Announced and implemented on 13 April 2018 by CMA CGM, the surcharge update reduced the rate on cargo from all ports worldwide to Aden (Yemen) by $200 USD per FEU from the previous level[5] (CMA CGM Notice). The event is classified as a war-related disruption.

Background.— In January 2018, the Battle of Aden between the Southern Transitional Council and Hadi government forces had caused severe port congestion and prompted CMA CGM to impose a $600 per container surcharge (see the Battle of Aden event, January 2018). By February 2018, the STC had returned territorial control to government forces, and the immediate security situation around the port had stabilized (Battle of Aden, Wikipedia). Meanwhile, the humanitarian crisis in Yemen had been designated the worst in the world, with over 22 million people requiring assistance (UN Press Release).

Cause of the price surcharge.— On 3 April 2018, a High-Level Pledging Event for the Humanitarian Crisis in Yemen, co-hosted by the United Nations, Sweden, and Switzerland in Geneva, secured over USD 2 billion in donor commitments to support the 2018 Yemen

[5]The previous surcharge was $600 USD per FEU (CMA CGM 09-Jan-2018)

Humanitarian Response Plan (ReliefWeb). The pledges were directed at scaling up aid delivery, enhancing humanitarian access, and streamlining logistics at Yemen's ports, with Aden serving as the primary entry point (OCHA). The improved security conditions following the end of the Battle of Aden, combined with the international commitment to strengthen port logistics, led CMA CGM to reduce the Port Congestion Surcharge by $200 on 13 April 2018, ten days after the Geneva conference.

Tropical Storm and Monsoon Flooding Disrupt Yangon, Myanmar (June 2018)

Technical details of the surcharge.— Announced and implemented on 1 June 2018 by CMA CGM, the surcharge applied to all imports into Myanmar ports at a rate of $200 USD per container (CMA CGM Notice). The event is classified as a weather-related disruption.

Background.— The Port of Yangon is Myanmar's dominant commercial gateway, handling over ninety per cent of the country's trade. Unlike a deep-water coastal port, Yangon is a river port situated along the Yangon River (Skuld, Yangon Port Information Sheet), whose approach is obstructed by inner and outer sand bars. Depending on the tide and season, these bars limit the maximum vessel draft effectively barring standard container ships and forcing carriers to rely on small feeder vessels (Logistics Cluster).

Cause of the price surcharge.— The surcharge reflected three distinct supply-side shocks. First, on 29 May 2018, an unanticipated tropical storm made landfall on the coast of Rakhine State, bringing strong winds and heavy rains to nine states and regions, including Yangon, where more than 750 houses and other buildings were damaged (ReliefWeb). The storm triggered localized flooding and landslides across the Yangon hinterland, disrupting the inland road corridors feeding the port even before the monsoon had set in. Second, and partly as a consequence of the resulting siltation of the navigation channel, an unexpected tightening of the permissible draft at the Yangon River's inner and outer bars sharply curtailed the intake capacity of every feeder service calling at the port, with industry sources reporting a massive container jam around Yangon and reduced yard productivity (Journal of Commerce). The

constraint proved severe enough that in July 2018, Hapag-Lloyd ceased accepting bookings for Yangon imports entirely, citing the draft limitations (Hapag-Lloyd Advisory). Third, the onset of the 2018 southwest monsoon in June brought an exceptionally heavy rainy season: rains had affected ten states and regions by mid-June, and by 30 July at least 16 people had died and roughly 268,000 had been temporarily evacuated across seven states and regions, with flood warnings issued by the Department of Meteorology and Hydrology (OCHA, FloodList, ReliefWeb). The flooding and landslides damaged inland transport corridors and disrupted truck movements to and from the port, preventing the yard backlog from clearing through normal land-side dispatch. CMA CGM issued the Port Congestion Surcharge on 1 June 2018, a shock driven by the sequential interaction of a pre-monsoon tropical storm, an unanticipated tightening of navigational constraints at a river port, and an exceptionally heavy monsoon season, entirely unrelated to any change in Myanmar's import demand.

Low Water Level St. Laurent Seaway, Canada (July 2018)

Technical details of the surcharge.— Announced on 16 July 2018 and implemented on 1 August 2018 by CMA CGM, the surcharge applied to all cargo from North Europe to Canada East Coast on the St Laurent service at a rate of $150 USD per container (CMA CGM Notice). The event is classified as a weather-related disruption.

Background.— The St. Laurent River is the sole navigable waterway connecting the Atlantic Ocean to the Port of Montreal, Canada's second-largest container port and the primary destination for European containerized trade with eastern Canada (Port of Montreal, Wikipedia). Outflows from Lake Ontario into the St. Laurent are regulated by the International Lake Ontario–St. Laurent River Board. In 2017, record-high water levels on Lake Ontario, driven by exceptional precipitation and record inflows from the Ottawa River, forced the Board to release the highest sustained outflows ever recorded from mid-June into August, far above normal operating ranges (International Joint Commission, FAQ). To protect upstream shoreline communities from continued flooding, the Board maintained these elevated releases well

into the autumn and through the winter of 2017–2018.

Cause of the price surcharge.— The prolonged high outflows drew down Lake Ontario's reserves. By the summer of 2018, the downstream effect materialized: while Lake Ontario itself returned to near-average levels, the St. Laurent navigation channel experienced very low, and at times record-low, water levels, with severe impacts on commercial navigation (International Joint Commission, Outflows 2018–2019). The reduced water depth forced vessels on the St. Laurent service to operate with lighter cargo loads and reduced speeds to avoid grounding, diminishing per-voyage capacity and raising unit costs. CMA CGM issued the Low Water Surcharge on 16 July 2018, a shock to shipping capacity driven by an unanticipated hydrological reversal from the 2017 flooding, entirely unrelated to any change in Canadian import demand.

Port Worker Strikes Disrupt Shipping in Portugal (September 2018)

Technical details of the surcharge.— Announced on 10 September 2018 and implemented on 18 October 2018 by CMA CGM, the surcharge applied to exports from Lisbon (Portugal) at a rate of $150 USD per container (CMA CGM Notice). The event is classified as a strike-related disruption.

Background.— Portugal's port labor system had been undergoing liberalization since the government adopted a new port law that entered into force on 1 February 2013, initiating a process that progressively deteriorated conditions for dockworkers, including the dismissal of 47 workers in Lisbon without reason and, according to the European Transport Workers' Federation, a breach of ILO Convention 137 on dock work which Portugal has ratified (ITF Global). The reform loosened the traditional stevedoring employment model by redefining which jobs belonged to the union labor pool, enabling the hiring of casual, non-union workers through staffing agencies (Labor Notes). A previous wave of strikes in 2012–2013 caused the port of Lisbon to lose approximately 40% of its average cargo throughput, and a further dispute beginning in November 2015 centered on the unions' accusation that port operators

were training non-unionized labor as preparation for replacing the existing unionized workforce (Lexology). The Sindicato dos Estivadores e da Actividade Logística (SEAL), the main stevedores' union, remained the principal opponent of these reforms throughout the period.

Cause of the price surcharge.— The immediate trigger came from a legal opinion issued by the Mobility Institute, the national body coordinating transport under the Ministry of Economics, which ruled that weekend and holiday work did not constitute overtime, forcing dockworkers, particularly at Leixões, to exceed the annual legal limit of 250 overtime hours (Socialist Alternative). In response, SEAL called a nationwide dockers' strike on 27 July 2018 with reportedly 100% participation, followed by a four-week overtime ban from 13 August to 10 September across all eight Portuguese ports where the union was represented, Leixões, Figueira da Foz, Lisbon, Setúbal, Sines, Caniçal, Ponta Delgada, and Praia da Vitória, citing the *"increasing proliferation of anti-union practices"*, particularly severe at Leixões and Caniçal (The Portugal News). The union also framed the action as a response to retaliation against workers joining SEAL and the systematic use of precarious casual contracts paying less than half the wage of regular workers (Guilhotina). The labor action progressively disrupted vessel berthing schedules and cargo throughput, with Port Strategy reporting that the strikes were *"slowly but surely beginning to have an impact"* on port traffic at Portugal's seven principal ports (Port Strategy), and drew international solidarity, with dockers from Spain, Belgium, Denmark, and Slovenia joining demonstrations in Lisbon (Guilhotina). CMA CGM issued the surcharge on 10 September 2018, a shock to shipping capacity driven by a labor dispute over employment terms and anti-union practices, entirely unrelated to any change in Portuguese export demand.

Shuwaikh Congestion from Dust Storms (September 2018)

Technical details of the surcharge.— Announced on 25 September 2018 and implemented on 25 October 2018 by CMA CGM, the surcharge applied to imports at Shuwaikh (Kuwait) at a rate of $300 USD per container (CMA CGM Notice). The event is classified as a

weather-related disruption.

Background.— Shuwaikh Port had been subject to recurring congestion surcharges since 2015, driven by a combination of customs bottlenecks, road-construction disruptions, and chronic infrastructure constraints. The port's storage envelope is fixed and limited, comprising a finite number of closed warehouses and an open storage yard whose capacity has long been recognised as insufficient for sustained throughput. Truck movements to and from the port are further constrained by a municipal regulation banning trucks from the inner city during weekday rush hours, restricting empty-container returns to off-peak windows.

Cause of the price surcharge.— Between June and September 2018, Kuwait experienced an exceptionally severe sequence of sand and dust storms, with wind speeds reaching 70 km/hour and visibility falling below 300 metres. The storms forced the closure of three Kuwaiti ports and the disruption of air traffic, and the volume of sand removed from desert highways rose from a monthly average of 30,000 m^3 to roughly 200,000 m^3 during the June–August stretch (Al-Hemoud et al., 2020). The repeated suspensions of maritime traffic at Shuwaikh, layered on top of the port's already strained storage envelope, produced an acute backlog: by late September warehouses had filled to their physical limit, the open yard ran out of space for incoming containers, and even off-port supplier warehouses reached saturation, leaving carriers without anywhere to dispatch discharged boxes. Within a single week, four major carriers responded by imposing simultaneous congestion surcharges, signalling a coordinated industry recognition of the severity of the disruption: CMA CGM announced its surcharge on 25 September; Hapag-Lloyd applied its congestion surcharge effective the same day (Container News); ANL re-implemented a port congestion surcharge effective 15 October citing berth congestion resulting in service disruptions, operational delays and increased operating costs (ANL Customer Advisory); and MSC followed in October with its own Shuwaikh port congestion surcharge (MSC Customer Advisory).

Aden Port Congestion from Yemen Conflict (November 2018)

Technical details of the surcharge.— Announced on 28 November 2018 and implemented on 1 December 2018 by CMA CGM, the surcharge applied to cargo from all ports worldwide to Aden (Yemen) at a rate of $800 USD per FEU, raising the previously applicable rate by $400 USD per FEU (CMA CGM Notice). The event is classified as a war-related disruption.

Background.— Since June 2018, the Saudi-led coalition had been conducting a major offensive against Houthi-held Hodeidah in an effort to sever supply lines to the north (Battle of Al Hudaydah, Wikipedia). With Hodeidah's port infrastructure already degraded by 2015 airstrikes (see Section 6) and its approaches an active combat zone, Aden, a smaller facility on the Gulf of Aden, remained the only viable commercial gateway for southern and central Yemen, and had been operating under elevated congestion surcharges since January 2018 following the Battle of Aden.

Cause of the price surcharge.— On 1 November 2018, the coalition deployed approximately 10,000 additional troops and launched a renewed offensive on Hodeidah, with over 200 airstrikes reported around the city and intense fighting at the airport and university (Battle of Al Hudaydah, Médecins Sans Frontières). Between 3 and 12 November alone, more than 200 air attacks hit the city, and at least 600 Houthi fighters were killed (Al Jazeera). As commercial traffic through Hodeidah became impracticable, cargo flows diverted to Aden: the World Food Programme reported in mid-November 2018 that only one containerised cargo vessel had received coalition clearance to berth at Hodeidah since the end of the November 2017 blockade (WFP Yemen Situation Report), sharply worsening berth congestion and waiting times. CMA CGM accordingly raised its existing Aden Port Congestion Surcharge on 28 November 2018, a shock driven by the November escalation of the Hodeidah offensive, entirely unrelated to any change in Yemen's import demand.

Regulatory Shock Across China, Hong Kong, and Taiwan (December 2018)

Technical details of the surcharge.— Announced on early December 2018 and implemented on 1 January 2019 by CMA CGM and MSC, the surcharge applied to cargo from and to all China, Hong Kong SAR, and Taiwan ROC ports at an average rate of $41.25 USD per container (CMA CGM Notice, MSC Notice). The event is classified as an operational disruption.

Background.— Following the 1 October 2018 enforcement of the 0.50% sulphur cap in the Yangtze River Delta, the industry expected only a gradual extension consistent with China's 2015 implementation plan, under which the cap would apply at berth in all DECA ports from 2018 and while sailing within the existing three DECAs from 1 January 2019 (EGCSA, Riviera Maritime Media). As late as 7 November 2018, marine insurer Gard advised that, although media reports had suggested China might extend the cap to its entire territorial sea, *"Gard is not aware of formal notice to this effect"* (Gard, "Sulphur cap ahead!").

Cause of the price surcharge.— The regulatory landscape changed abruptly in late November 2018. On 30 November 2018, China's Ministry of Transport issued a new *Marine Air Emission Control Areas Implementation Scheme* that designated China's entire territorial sea as a Coastal Emission Control Area, plus a new Hainan Coastal ECA and two Inland ECAs covering parts of the Yangtze and Xi Jiang rivers, mandating 0.50% sulphur fuel for all vessels operating within these zones from 1 January 2019, a far broader scope than the three pre-existing DECAs (Manifold Times; Wang and Peng (2019)). The cost gap between compliant and non-compliant fuels also induced a documented evasion response: empirical work using AIS data shows that some carriers altered their sailing patterns to minimise time spent inside the Chinese ECA boundary, sailing the cheapest fuel outside the zone and switching to compliant fuel only upon entry, which extended sailing distances and raised total operating costs across the Europe–China trade lane (Tan et al. (2021), Weng et al. (2022)). Two weeks after China's MoT notice, CMA CGM extended its surcharge to all Chinese,

Hong Kong, and Taiwanese ports (CMA CGM Notice), and MSC followed on 17 December with an identical expansion (MSC Customer Advisory). The price represents a shock to shipping capacity driven by the unanticipated late-November expansion of China's sulphur ECA to cover its entire territorial sea, compounded by concurrent Hong Kong and Taiwan regulations, entirely unrelated to any change in East Asian trade demand.

Low Water Level St. Laurent Seaway, Canada (December 2018)

Technical details of the surcharge.— Announced on 20 December 2018 and implemented on 17 January 2019 by CMA CGM, the surcharge applied to all cargo from North Europe to Canada East Coast on the St Laurent service at a rate of $125 USD per container (CMA CGM Notice). The event is classified as a weather-related disruption.

Background.— In July 2018, CMA CGM had already imposed a Low Water Surcharge on the same route after the record-high Lake Ontario outflows of 2017 produced very low water levels on the St. Laurent navigation channel by the summer of 2018. The surcharge was framed as a response to a transient hydrological reversal, with Lake Ontario levels returning toward the long-term average through the autumn. By late 2018 the immediate summer crisis had abated, and shippers on the Europe–Canada trade had no reason to expect a second, winter-specific disruption of equal severity.

Cause of the price surcharge.— A new and distinct hydrological shock materialised in the second half of December 2018. In its Water Level Forecasts Bulletin dated 17 December 2018, three days before CMA CGM's surcharge announcement, the Canadian Coast Guard warned of "an important decrease of water levels" over the mid-term, driven by upstream control-structure management prioritising the formation of stable ice covers for the winter season (Canadian Coast Guard, Water Level Forecasts Bulletin, 17 December 2018). The Bulletin projected expected lowest water levels at Montreal collapsing from 0.80 m above chart datum in the 18–20 December window to just 0.10 m by the 1–7 January window, a reduction of roughly 70 cm within three weeks that far exceeded the normal seasonal drawdown. Each

10 cm reduction in channel depth forces approximately 3,000 metric tonnes of cargo to be removed from a freighter (Globe and Mail), so the forecast drawdown implied a load reduction on the order of 20,000 tonnes per deep-draft vessel calling Montreal in the first weeks of 2019. The trigger was therefore neither a continuation of the summer condition nor a predictable seasonal pattern, but an operational decision by the International Lake Ontario–St. Laurent River Board and downstream dam operators to suppress outflows in preparation for ice-cover formation, superimposed on a basin already depleted by the record outflows of 2017. CMA CGM issued the second Low Water Surcharge on 20 December 2018 at a rate of $125 per container, a shock driven by an unanticipated winter-regime drawdown of the St. Laurent navigation channel, entirely unrelated to any change in Canadian import demand.

Gulf Shipping Disruptions from Strait of Hormuz Attack (July 2019)

Technical details of the surcharge.— Announced on July 2019 and implemented on 1 August 2019 by CMA CGM and MSC, the surcharge applied to cargo from and to Bahrain, Iraq, Kuwait, Oman, Qatar, Saudi Arabia, and the UAE at an average rate of $76 USD per container (CMA CGM Notice, MSC Notice). The event is classified as a war-related disruption.

Background.— The Strait of Hormuz is the narrow waterway connecting the Persian Gulf to the Gulf of Oman, through which essentially all commercial shipping to and from the Gulf states must transit. Since the end of the 1980s Tanker War, the strait had seen no direct strikes on merchant shipping, and war-risk insurance premiums for Gulf voyages had settled to routine low levels on the order of 0.02–0.05% of hull value (Euronews). Tensions escalated sharply in May 2019 following the U.S. withdrawal from the Iran nuclear deal and the reimposition of sanctions on Iranian oil exports.

Cause of the price surcharge.— In a period of less than three months, tensions in the Gulf escalated into near-open conflict through a rapid sequence of attacks and vessel seizures that

severely disrupted commercial shipping operations across the entire region. On 12 May 2019, four commercial ships were damaged in an unclaimed explosion near the UAE port of Fujairah, the first direct strikes on Gulf merchant shipping in over three decades, forcing carriers to reassess route safety and causing immediate operational uncertainty. On 13 June 2019, two oil tankers were attacked while transiting the Gulf of Oman and both caught fire, compelling vessels to alter transit procedures and driving insurance premiums sharply higher. On 18 July 2019, Iran's Islamic Revolutionary Guard Corps (IRGC) seized an oil tanker in the Strait of Hormuz without disclosing her nationality. The following day, 19 July 2019, the IRGC impounded the British-flagged oil tanker *Stena Impero*, demonstrating that flag state offered no protection against seizure and forcing carriers to consider armed escorts and alternative routing. On 31 July 2019, a foreign oil tanker was seized in the Gulf off the Iranian coast, reportedly the third foreign vessel seized by Iran in two weeks (China P&I Club, LP 31/2019). The cumulative effect was a fundamental deterioration in the operating environment for all Gulf-bound shipping: the London insurance market's Joint War Committee issued JWLA-024, formally adding Oman, the UAE, and the Persian Gulf west of longitude 58°E to the Hull War, Piracy, Terrorism and Related Perils Listed Areas (Latitude Brokers, Maritime Cyprus, BIMCO), triggering mandatory additional premiums, vessel rerouting, and disruptions to schedules across all Gulf trade lanes. CMA CGM issued the War Risk Surcharge on 2 July 2019, a shock to shipping capacity driven by the sudden collapse of operational security across the Gulf corridor, entirely unrelated to any change in trade demand on the affected routes.

UAE Military Drawdown Triggers Congestion at Aden (July 2019)

Technical details of the surcharge.— Announced on 19 July 2019 and implemented on 20 August 2019 by CMA CGM, the surcharge applied to cargo from all origins to Aden, Yemen, at a rate of $300 USD per container (CMA CGM Notice). The event is classified as a war-related disruption.

Background.— Aden had been under persistent conflict-related congestion since the Saudi-led coalition's intervention in 2015, with the port serving as the only viable commercial gateway for southern Yemen. The January 2018 Battle of Aden between the STC and Hadi-government forces had severely disrupted port operations, and the November 2018 Hodeidah offensive had further strained Red Sea logistics networks. From 2015 through mid-2019, security at Aden's port and its landside supply corridor had been underwritten in large part by the United Arab Emirates, which operated the dominant ground force in the coalition's southern sector. The UAE had led Operation Golden Arrow in July 2015 to recapture Aden from the Houthis and had maintained advisory, logistical, and direct combat units in and around the city thereafter, protecting the commercial corridor through which essentially all humanitarian and commercial cargo to southern Yemen transited (Middle East Institute).

Cause of the price surcharge.— On 8 July 2019, the UAE publicly announced a major drawdown of its military forces from Yemen, shifting from a military-first to a peace-first strategy and initiating a phased withdrawal that had already moved more than half of its contingent out of the country, including from the southern port of Aden (Reuters via Euronews). The announcement specified a reduction to roughly 4,000 soldiers and the removal of heavy weapons, including tanks, artillery, and anti-missile systems (Middle East Institute, 11 July 2019). Commentary published within days of the announcement noted explicitly that the drawdown risked leaving a security vacuum at Aden and raised the prospect of open confrontation between rival factions, the Hadi government, the STC, and UAE-trained forces, including the Security Belt Forces, that the UAE had until then kept in check (Foreign Policy). For CMA CGM's Aden service, the drawdown represented a step-change in the security environment of the port's immediate hinterland: the exit of the coalition's dominant ground force implied higher war-risk exposure, degraded landside security for cargo flows, and elevated probability of renewed urban combat around the port facilities. Eleven days after the UAE announcement, CMA CGM issued the Port Congestion Surcharge on 19 July 2019 at $300 per container, with implementation on 20 August 2019. The subsequent

sequence of events confirmed the deteriorating trajectory that the surcharge had priced in: on 1 August 2019 a Houthi-claimed missile attack on a military parade in Aden killed dozens of security personnel, and by 10 August STC separatists had seized the presidential palace and taken control of most of the city in open combat against Hadi-government forces, with Riyadh accusing the UAE of backing a *"coup"* (Al Jazeera, 29 August 2019). The surcharge represents a shock to shipping capacity driven by the sudden withdrawal of the Gulf-state security architecture that had protected Aden's commercial operations, entirely unrelated to any change in Yemen's import demand.

Transport Strikes Disrupt Shipping in South France (January 2020)

Technical details of the surcharge.— Announced on 22 January 2020 and implemented on 1 February 2020 by Maersk, the surcharge applied to cargo from and to Southern France at a rate of $425 USD per container (Maersk Notice). The event is classified as a strike-related disruption.

Background.— On 5 December 2019, French workers launched a nationwide strike to protest President Macron's proposed pension reform (2019–2020 French Pension Reform Strike, Wikipedia). The strike became the longest in modern French history, paralysing transportation across the country: railway workers, public transport operators, truck drivers, and port workers joined the action, shutting down high-speed rail services, oil refineries, and inland logistics networks (Washington Post). Port workers' unions joined the nationwide movement from the outset, and by early January 2020 unions representing air traffic controllers, dockworkers, and refinery staff had all been drawn into the action, with strike-related blockades shutting refineries and disrupting maritime transport (Euronews).

Cause of the price surcharge.— In mid-January 2020 the strike action escalated sharply at the Port of Fos-sur-Mer, Maersk's primary southern French gateway. A 72-hour dockworkers' and port workers' strike was called from 14 January until 17 January, overlapping with a separate 120-hour towage-operations strike running from 15 January until 20 January,

which effectively halted vessel pilotage into and out of the port (Container News). On the multimodal side, all rail services into and out of Fos-sur-Mer were cancelled, forcing containers to be stored at inland depots and re-routed to truck transport where available (Maersk Customer Advisory). Six days after the end of the January dockworkers' strike, and with further industrial action threatened, Maersk issued the Local Charge on 22 January 2020 at $425 per container, with implementation on 1 February. The charge represents a shock to shipping capacity driven by a nationwide labour action over domestic pension reform that escalated into specific, concurrent dockworkers', tugboat, and rail strikes at Fos-sur-Mer, entirely unrelated to any change in French trade demand.

Port Congestion from Secure Workplace Protocols, Felixstowe (September 2020)

Technical details of the surcharge.— Announced on 22 September 2020 and implemented on 1 October 2020 by CMA CGM, the surcharge applied to cargo from and to Felixstowe to/from Far East and South East Asia at a rate of $300 USD per container (CMA CGM Notice). The event is classified as an operational disruption.

Background.— The Port of Felixstowe is the United Kingdom's largest container port, handling approximately 36 per cent of the country's containerised trade. Felixstowe had been operating under a persistent structural constraint since the June 2018 migration to Hutchison Ports' in-house nGen terminal operating system (Seatrade Maritime). UK import volumes had been running above normal since the easing of the first lockdown in June 2020 as retailers rebuilt inventories depleted during April–June, and this elevated volume had been absorbed through the summer without triggering carrier surcharges. By mid-September, however, several supply-side pressures were compounding the baseline constraint: working days had been lost to summer storms and UK bank holidays, typhoons in Asia were delaying vessel arrivals and concentrating ships into compressed berthing windows, a container-trucking driver shortage had emerged as overseas nationals returned home during lockdown and other workers remained on furlough, and empty containers were accumulating on the

quay faster than they could be processed out (PFE Express, 18 September 2020; Offshore Energy, 23 September 2020).

Cause of the price surcharge.— The triggering cause of the September surcharge was a sharp tightening of COVID-secure workplace protocols that materially worsened labour availability at an already-strained terminal. On 21 September 2020 the four UK Chief Medical Officers raised the national COVID-19 alert level from 3 to 4, citing exponentially rising transmission (GOV.UK, 21 September 2020). The following day, the government announced a new package of COVID-secure workplace measures, and on 28 September the Health Protection (Self-Isolation) Regulations 2020 came into force, converting self-isolation from government advice into a legally enforceable duty backed by fines and creating a new criminal offence for employers who knowingly allowed self-isolating workers onto the premises (UK Statutory Instruments 2020/1045). For a labour-intensive terminal that already required mandatory deep cleaning between every shift changeover and was operating with reduced staff, the legal enforcement of self-isolation removed any residual discretion over whether exposed workers could attend the workplace, further compressing the effective labour force. CMA CGM's own surcharge notice identified the causal mechanism explicitly, attributing the cost increase to *"reduced terminal productivity due to Covid-safe working practices and the deep cleaning required at each shift changeover, and reduced driver availability in the container sector"*. CMA CGM issued the surcharge on 22 September 2020, the same day as the UK government's announcement, a shock to shipping capacity driven by pandemic-imposed labour and productivity constraints, entirely unrelated to any change in UK import demand.

Australia Port Congestions from Strikes (September 2020)

Technical details of the surcharge.— Announced in September 2020 and implemented from October 2020 by CMA CGM, Maersk, and MSC, the surcharge applied to cargo from and to Sydney, Melbourne, and Brisbane at a rate of $556.7 USD per container (CMA CGM Notice, MSC Notice, Maersk Notice). The event is classified as a strike-related disruption.

Background.— Australia's east-coast container trade is concentrated at three terminals, Port Botany (Sydney), the Port of Melbourne, and the Port of Brisbane, which together handle the majority of the country's containerised imports. Patrick Terminals is Australia's largest stevedore, operating facilities at all three east-coast ports plus Fremantle in Western Australia. Through the first half of 2020, Patrick had been in enterprise-agreement negotiations with the Maritime Union of Australia (MUA) covering its entire workforce; by late summer these talks had been running for seven months without resolution (Patrick Terminals Media Release).

Cause of the price surcharge.— On 4 September 2020, the MUA launched coordinated protected industrial action across all four Patrick facilities. At Port Botany the action comprised an immediate four-hour work stoppage, an indefinite ban on overtime, an indefinite ban on working on the Port Side Rail Project, and a ban on handling vessels subcontracted from other stevedores, with further escalation, including rolling one-hour stoppages, 24-hour strikes, and a 14-day continuous-action programme, scheduled through September (Dieterle-Victory Industrial Action Schedule). Within four weeks Patrick reported that productivity at Port Botany had been cut by 40 to 50 per cent, shipping schedules had slipped by nine days and were worsening by half a day for every day the action continued, and a backlog of approximately 35,000 containers had accumulated at the terminal (The Loadstar; Patrick Terminals Media Release). The industry-wide impact was immediate: Shipping Australia estimated the cost of a single day's vessel delay at A$25,000, and Patrick subsequently applied to the Fair Work Commission to terminate the industrial action on grounds of significant economic harm (Patrick Terminals Media Release). The first surcharge from CMA CGM, Maersk, and MSC was issued on 8 September 2020, four days after the action began at Port Botany, a shock to shipping capacity driven by protected industrial action over an unresolved enterprise agreement dispute at Australia's largest stevedore, entirely unrelated to any change in Australian import demand.

Bilbao Port Strike Escalation (November 2020)

Technical details of the surcharge.— Announced on 7 November 2020 and implemented on 16 November 2020 by CMA CGM and MSC, the surcharge applied to cargo from and to Bilbao at a rate of $250 USD per container (CMA CGM Notice, MSC Notice). The event is classified as a strike-related disruption.

Background.— The Port of Bilbao is the main container gateway for the Basque Country and one of the principal maritime trade nodes for northern Spain. Stevedoring operations at Bilbao are organised through the Port Employment Centre (Centro Portuario de Empleo, CPE), which hires approximately 320 permanent dockers supplemented by around 100 temporary workers. The collective agreement covering Bilbao stevedores expired in 2019, and negotiations between the employer and the stevedoring unions continued without resolution through 2020. The unions had accumulated grievances including "continuous breaches" of the expired agreement on rest days and workforce incorporation, deteriorating machinery conditions, and what they characterised as "bad faith" negotiation by the employer side (European Transport Workers' Federation).

Cause of the price surcharge.— A first phase of strike action ran from 9 to 25 October 2020 with partial and full stoppages on designated days, followed by a second phase from 26 October to 9 November with strikes applying to half of each working day (Ports Europe). Mediated negotiations collapsed in early November without agreement. In the first days of November the five stevedoring unions filed a new strike notice covering 9 November to 9 December 2020, crucially, this third phase escalated the action to full 24-hour stoppages every day of the month, rather than the partial stoppages of the earlier phases (Lamaignere News). The escalation also triggered solidarity strike notices at Valencia and Barcelona for 10, 13, 18, 20, 26, and 27 November, raising the prospect of disruption cascading across Spain's main container ports (Lamaignere News). Operational capacity at Bilbao fell to approximately 50 per cent, loading and unloading operations normally completed in hours stretched

to days, and carriers were forced to reroute vessels to Santander and Gijón (Allmed). CMA CGM and MSC issued the surcharge on 7 November 2020, two days before the escalated third phase was due to begin, a shock to shipping capacity driven by the sudden escalation of an unresolved collective-bargaining dispute from partial to full-day stoppages, entirely unrelated to any change in Spanish import demand.

Automated Straddle Failures Disrupt Auckland (November 2020)

Technical details of the surcharges.— Multiple carriers issued congestion surcharges on cargo to and from Auckland in the first half of November 2020, for an average surcharge of $476 USD per FEU (CMA CGM Notice, Maersk Notice, MSC Notice). All events are classified as operational disruptions.

Background.— The Port of Auckland is New Zealand's principal import gateway, historically handling roughly one-third of the country's seaborne trade and operating as the first port of call for a large share of vessels visiting the country (International Transport Workers' Federation, 2023). The Fergusson Container Terminal handles essentially all of the port's containerised throughput within a fixed urban footprint that cannot be expanded by reclamation, with the operational consequence that any reduction in usable yard area or in straddle-carrier availability translates directly into berth and gate congestion. From 2016, Ports of Auckland Limited (POAL) had embarked on a partial automation programme intended to replace manual straddle carriers with 27 automated units, an undertaking POAL itself described as a world-first attempt to automate a functioning container port (POAL Annual Report 2020).

Cause of the surcharges.— The November surcharges were triggered by the abrupt breakdown of the automation programme during its first months of live operation, an event entirely outside the operational expectations of carriers and shippers serving the New Zealand trade. In early November 2020, software faults forced POAL to take the entire automated straddle fleet offline for an urgent fix (ITF, 2023), and shortly thereafter one of the automated units

became overloaded, failed to negotiate a turn, toppled over a concrete retaining barrier and crashed into a stack of containers (Radio New Zealand, 1 December 2020). Because the terminal had been physically partitioned into automated and manual yards, each suspension of the automated fleet removed approximately one-third of the port's effective handling capacity at a stroke and concentrated all residual throughput onto the manual yard, which had simultaneously been thinned by the recruitment freeze and was running with critical labour shortages across crane drivers, straddle operators and lashers (Radio New Zealand, 18 November 2020). The combined effect was an unprecedented compression of berth availability: vessel waiting times rose to 10–13 days, fixed berthing windows were suspended, and carriers were forced into port omissions and route restructurings to recover schedule integrity across the wider Oceania network (Container News, 4 November 2020). MSC issued the first congestion surcharge on 3 November 2020, with CMA CGM and Maersk following within the same week, each citing the elevated operational costs imposed by extended berth waits, transhipment to alternative ports, empty-container repositioning and lost sailings (New Zealand Herald, March 2022). Notably, no comparable surcharge was levied at any other New Zealand port over the same window despite shared exposure to pandemic-era global supply-chain stress, isolating the Auckland-specific automation breakdown as the proximate cause (ITF, 2023). The surcharges represented a shock to shipping capacity driven by the unanticipated failure of an in-progress automation conversion.

Lockdown-Induced Congestions Across UK Container Gateways (November 2020)

Technical details of the surcharge.— Announced on 19 November 2020 and implemented on 1 December 2020 by CMA CGM, the surcharge applied to cargo from and to all UK ports to/from Asia at a rate of $240 USD per container (CMA CGM Notice). The event is classified as an operational disruption.

Background.— UK containerised trade is concentrated in three gateways: Felixstowe (approximately 36 per cent of UK container volume), the Port of Southampton, and DP World's

London Gateway. Through summer and early autumn 2020, operational pressure had been building specifically at Felixstowe, driven by the combination of COVID-secure workplace protocols and an unresolved legacy of the failed 2018 migration to the nGen terminal operating system. On 22 September 2020 CMA CGM had issued a $300 per container surcharge covering Felixstowe-Asia lanes. In the first weeks of November, the congestion concentrated at Felixstowe spilled decisively across the UK container network. Shippers and carriers diverting away from Felixstowe had overwhelmed Southampton and London Gateway, which were absorbing volumes for which they had no spare capacity.

Cause of the price surcharge.— The cause of the November surcharge was a sharp tightening of the regulatory environment for terminal labour across the UK. On 31 October 2020 the UK government announced a second national lockdown for England, which took legal effect on 5 November and reimposed workplace restrictions that had been relaxed over the summer (House of Commons Library). For UK container gateways, the second lockdown compounded the COVID-secure workplace protocols already in force, mandatory deep cleaning between shift changeovers, extended self-isolation requirements, and restrictions on workforce density, further compressing terminal productivity just as the three main gateways were absorbing elevated divert-and-reroute volumes from the Felixstowe crisis. Reuters Events reported that UK ports have faced major disruption throughout 2020, but this has compounded significantly towards the end of the year, with the second lockdown and associated COVID measures materially worsening the operational position of terminals already running above their pandemic-constrained capacity (Reuters Events). CMA CGM issued the UK-wide Port Congestion Surcharge on 19 November 2020, extending its October surcharge regime from Southampton alone to cover Felixstowe, Southampton, and London Gateway together, at $240 per container effective 1 December (CMA CGM; Container News). The surcharge represents a shock to shipping capacity driven by the second national lockdown's tightening of workplace restrictions on already-stressed UK container terminals, entirely unrelated to any change in UK import demand.

Empty-Return Refusal and Chassis Gridlock at LA–LB Ports (November 2020)

Technical details of the surcharge.— Announced on 4 November 2020 and implemented on 4 December 2020 by MSC, the Emergency Intermodal Charge (EIC) applied to dry containers moving on carrier haulage, door service, or carrier-arranged drayage between the Far East and US inland destinations via Los Angeles and Long Beach, at a rate of $350 USD per container (MSC Notice). Shipments using a Client Nominated Trucker were exempt, confining the charge to carrier-managed inland services. The event is classified as an operational disruption.

Background.— The Ports of Los Angeles and Long Beach together form the largest container gateway in the United States and the primary Pacific-coast entry point for East Asian imports, with landside cargo cleared through a combination of drayage to the Inland Empire warehouses and intermodal rail to destinations such as Chicago, Memphis, and Dallas. The hinterland draws on a shared chassis infrastructure, covering approximately 62,000 chassis, of which roughly 13 per cent are typically out-of-service at any time for maintenance (O'Neill Logistics). The empty-return appointment system had been introduced to manage chassis recirculation and was designed to facilitate dual transactions where truckers could drop off an empty container and pick up a loaded one in a single terminal visit (MoreThanShipping). Supply-side capacity constraints had been building through 2020: Tropical Storm Isaias struck the East Coast on 3–4 August 2020, causing over 6.8 million power outages from South Carolina to Maine and damaging warehouse infrastructure across the Mid-Atlantic distribution corridor, reducing East Coast logistics capacity (Yale Climate Connections). Import volumes had been recovering through the summer and early fall as retailers began restocking inventory for the holiday season (Laufer Group).

Cause of the price surcharge.— The November surcharge was triggered by the unexpected collapse of the empty-container return system at the San Pedro Bay port complex in October 2020. Terminal operators began refusing truckers at scheduled empty-return appoint-

ments, citing slot unavailability for periods of up to three days; in response, drivers defensively reserved multiple appointment windows simultaneously, leaving approximately one-third of all scheduled slots unused and trapping chassis with empty containers off-terminal (Port X Logistics). The resulting chassis immobilisation sharply curtailed the recirculation of equipment through the port, creating a self-reinforcing shortage that industry observers described as reducing steamship lines, truckers, shippers, and intermediaries alike to a state of operational paralysis (MoreThanShipping). Terminal policies requiring single-transaction appointments further impeded efficiency, as carriers requiring combined import-discharge and empty-return movements in a single truck run were forced to compete for scarce slots while chassis sat idle (Supply Chain Dive). The ocean-carrier alliance structure compounded the disruption by preventing carriers from communicating empty-return locations in advance, transforming chassis repositioning into a logistically uncoordinated exercise (Supply Chain Dive). In early November 2020, the Harbor Trucking Association and California Trucking Association issued a joint statement warning that cargo flow at the Southern California ports was approaching complete gridlock, a declaration that prompted the Federal Maritime Commission to open a formal investigation into whether carrier practices on container return restrictions and detention charges posed a systemic risk to US trade capacity (Truckinginfo). Because MSC bore inland haulage and repositioning costs on carrier-managed shipments, the appointment system failure translated directly into elevated per-container operating costs on precisely the moves covered by the Emergency Intermodal Charge, while shipper-arranged inland transport remained unaffected. MSC issued the surcharge on 4 November 2020, a shock to carrier-managed inland shipping capacity driven by the October 2020 breakdown of the empty-return appointment system at San Pedro Bay, entirely unrelated to any change in underlying US import demand.

Low Water Level St. Laurent Seaway, Canada (January 2021)

Technical details of the surcharge.— Announced on 11 January 2021 and implemented on 7 February 2021 by CMA CGM, the surcharge applied to cargo from and to Canada East Coast at a rate of $50 USD per container (CMA CGM Notice). The event is classified as a weather-related disruption.

Background.— The St. Laurent River navigation channel is the sole deep-water marine corridor connecting the Atlantic Ocean to the Port of Montreal and the upstream St. Laurent Seaway, handling approximately 40 million tonnes of cargo annually and serving as the primary gateway for containerised trade to and from eastern Canada (Wikipedia, St. Lawrence Seaway). Navigation on the channel is highly sensitive to ambient water levels: between Quebec City and Montreal, the St. Laurent Seaway Management Corporation sets authorised maximum draft on a rolling basis according to prevailing water conditions, and when levels fall below navigation thresholds it imposes mandatory draft reductions and speed restrictions on deep-draft vessels, requiring carriers to lighten cargo loads and accept longer transit times to avoid grounding (Clear Seas, 2020).

Cause of the price surcharge.— The cause of the January surcharge was the International Lake Ontario–St. Laurent River Board's implementation of an above-Plan outflow deviation strategy effective 1 January 2021, triggered by a December 2020 hydrological risk assessment identifying persistent dry conditions across the Great Lakes Basin and forecasting below-average water levels in the St. Laurent navigation corridor through the winter season (International Joint Commission, 11 December 2020). The Board received authority from the International Joint Commission to deviate from Plan 2014 prescribed flows from 1 January through the end of February, implementing the third-highest outflows on record for that period, with its own forecasts projecting that Lake Ontario levels would remain below the long-term average through the season (International Joint Commission, March 2021). The resulting depressed water levels in the lower St. Laurent required the Seaway Management

Corporation to implement draft restrictions and speed reductions on deep-draft vessels approaching Montreal, directly raising the per-voyage operating costs borne by carriers on the Canada East Coast trade (LEVELnews, January 2021). CMA CGM issued the surcharge on 11 January 2021, a shock to shipping capacity driven by the January hydrological regime and associated navigation constraints on the St. Laurent, entirely unrelated to any change in Canadian import demand.

UK Surcharge on Indian Imports Due to Colombo Transshipment Breakdown (March 2021)

Technical details of the surcharge.— Announced on 12 March 2021 and implemented on 25 March 2021 by CMA CGM, the surcharge applied to cargo from India to Southampton and London Gateway at a rate of $400 USD per container (CMA CGM Notice). The event is classified as an operational disruption.

Background.— Colombo Port in Sri Lanka serves as the principal transshipment hub for CMA CGM's India–United Kingdom services, with feeder vessels collecting containers from Indian subcontinent ports, principally Chennai, Cochin, and Nhava Sheva, and transferring them at Colombo onto mainline vessels on the Europe-bound leg that call directly at Southampton and London Gateway (Seatrade Maritime). The port operates across three physically separate terminals, the Colombo International Container Terminal handling mainline calls and the Jaya Container Terminal and South Asia Gateway Terminal handling feeder operations, with all transshipment cargo requiring inter-terminal transfer by a dedicated fleet of prime movers between each feeder arrival and mainline departure (The Loadstar, November 2020). India accounts for approximately 61 per cent of Colombo's total transshipment volume, making the port's inter-terminal transfer capacity structurally critical to the performance of CMA CGM's India-to-UK feeder-to-mainline connection (Hellenic Shipping News, December 2020).

Cause of the price surcharge.— The cause of the March surcharge was a deliberate carrier

decision to withdraw mainline vessel and container capacity from the Asia–Europe trade lane and redeploy it to the transpacific. This reallocation directly removed scheduled mainline vessel calls from the Colombo–UK leg, severing the feeder-to-mainline connection on which all India-origin cargo for Southampton and London Gateway depends (PFE Express, 8 March 2021). With mainline departures reduced, feeder vessels arriving at Colombo from Indian ports were forced to wait over a week for an available berth, the feeder network fell to approximately 40 per cent of its normal operating capacity, and a backlog of over 10,000 containers from the Indian subcontinent accumulated at the port unable to depart (PFE Express, 8 March 2021). CMA CGM issued the surcharge on 12 March 2021, a shock to carrier-managed capacity on the India–UK route driven by the exogenous withdrawal of mainline vessel supply from the Colombo–Europe corridor, entirely unrelated to any change in UK import demand.

Suez Canal Blockage Surcharge and Extensions (May–August 2021)

Technical details of the surcharge.— The surcharge was announced and first implemented on 7 May 2021 by Maersk, applied to cargo from and to Europe and the Mediterranean, and extended across three subsequent monthly periods. The event is classified as an operational disruption.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
7 May 2021	7 May 2021	\$42.5	Maersk Notice
7 May 2021	1 June 2021	\$42.1	Maersk Notice
21 June 2021	1 July 2021	\$41.4	Maersk Notice
21 July 2021	1 August 2021	\$41.1	Maersk Notice

Background.— The Suez Canal is the sole artificial waterway connecting the Red Sea to the Mediterranean, providing the shortest maritime route between Asia and Europe and eliminating the need for vessels to circumnavigate the Cape of Good Hope, a detour of approximately 9,000 kilometres (Wikipedia, 2021 Suez Canal Obstruction). Approximately

12 per cent of global trade transits the canal annually, including a significant share of containerised cargo on the Asia–Europe trade lane, making it one of the most critical single points of failure in the global supply chain (Port Economics). The canal's single-lane configuration through its narrowest sections means that any obstruction of sufficient scale renders the entire waterway impassable in both directions simultaneously.

Cause of the price surcharge.— On 23 March 2021, the container ship Ever Given ran aground in the Suez Canal, blocking all transit for six days and halting an estimated 12 per cent of global trade (Wikipedia, 2021 Suez Canal Obstruction). Over 430 vessels queued at both entrances to the canal, tying up approximately 0.5 per cent of global shipping capacity per day of delay (Port Economics). When the canal reopened on 29 March, the compressed release of the queued fleet created a surge of simultaneous arrivals at European ports, overwhelming berth capacity and inland transport connections and generating a cascade of congestion that persisted well beyond the initial backlog clearance period (Port Economics). Maersk issued the surcharge on 7 May 2021 and extended it through August 2021, a shock to shipping capacity driven by the cascading operational aftermath of the Ever Given grounding, entirely unrelated to any change in European trade demand.

Low Water Level, St. Laurent Seaway, Canada (June 2021)

Technical details of the surcharge.— Announced in several notices from 1 June 2021 and implemented from 15 June 2021 by CMA CGM, Maersk, and MSC, the surcharge applied to cargo from Europe and the Mediterranean to Canada at an average rate of $266 USD per container (CMA CGM Notice, MSC Notice, Maersk Notice). The event is classified as a weather-related disruption.

Background.— The St. Laurent River navigation channel is the sole deep-water marine corridor connecting the Atlantic Ocean to the Port of Montreal and the upstream St. Laurent Seaway, handling approximately 40 million tonnes of cargo annually and serving as the primary gateway for containerised trade to and from eastern Canada (Wikipedia, St. Lawrence

Seaway). Navigation on the channel is highly sensitive to ambient water levels: the St. Laurent Seaway Management Corporation sets authorised maximum draft on a rolling basis according to prevailing water conditions, and when levels fall below navigation thresholds it imposes mandatory draft reductions and speed restrictions on deep-draft vessels, requiring carriers to lighten cargo loads and accept longer transit times (Clear Seas, 2020). The International Lake Ontario–St. Laurent River Board governs the water levels of the system under Plan 2014, which regulates outflows from Lake Ontario through the Moses-Saunders Power Dam; when lake levels fall below the low Criterion H14 threshold, the Board is authorised to reduce outflows below Plan-prescribed levels to preserve remaining water depth in the navigation corridor.

Cause of the price surcharge.— The cause of the June surcharge was the crossing of the critical low Criterion H14 threshold on 29 May 2021, which triggered the International Lake Ontario–St. Laurent River Board's emergency deviation authority after the preceding 12 months over Lake Ontario had been the driest since 1966 (International Joint Commission, 4 June 2021). Lake Ontario had risen only 2 cm in May against a seasonal average of 9 cm, leaving the lake 37 cm below its long-term average by the start of June and 64 cm below year-earlier levels, while water levels at Lake St. Laurent had already fallen below same-day historic lows on two separate occasions in May (LEVELnews, June 2021). In response, the Board reduced outflows at the Moses-Saunders Dam to 100 m^3/s below the Plan-prescribed Rule Curve on 29 May and to 200 m^3/s below on 5 June, directly constraining navigable draft across the approach to the Port of Montreal and forcing carriers to reduce cargo loads per transit (International Joint Commission, 4 June 2021). CMA CGM, Maersk, and MSC issued the surcharge in the first half of June 2021, a shock to shipping capacity driven by the May 2021 hydrological crisis and the resulting emergency navigation restrictions on the St. Laurent, entirely unrelated to any change in Canadian import demand.

Meishan Terminal Closure, Ningbo (August 2021)

Technical details of the surcharge.— Announced on 3 August 2021 and implemented on 15 August 2021 by MSC, the surcharge applied to all shipments from Asia to Europe and the Mediterranean at a rate of $1,000 USD per container (MSC Notice, Carson International). The event is classified as an operational disruption.

Background.— Ningbo-Zhoushan is the world's third-busiest container port by throughput and China's second-largest after Shanghai, handling a substantial share of containerised exports on the Asia–Europe trade lane. The port operates across multiple physically separate terminals, of which the Meishan Terminal is one of the primary deep-water container facilities, accounting for approximately 25 per cent of the port's total container handling capacity; a closure of Meishan alone is therefore sufficient to materially disrupt the port's throughput and the vessel schedules of all major carriers serving the Asia–Europe route (Fortune). China's national containment framework, applied uniformly across all port facilities, mandates the immediate suspension of all terminal operations upon the detection of a single positive case among the workforce, with no minimum threshold of cases required to trigger a full closure.

Cause of the price surcharge.— At the beginning of August, Chinese authorities invoked the national containment framework to shut down the Meishan Terminal in its entirety after a single dock worker tested positive, immediately removing approximately 25 per cent of Ningbo-Zhoushan's container capacity from service without prior warning to carriers (Al Jazeera). The policy-mandated closure was abrupt and non-negotiable: major shipping lines including Maersk, Hapag-Lloyd, and CMA CGM were compelled to reroute vessels and adjust schedules on short notice, while ships queued at anchor outside the terminal awaiting an uncertain reopening date (Seatrade Maritime). MSC issued the surcharge on 3 August 2021, a shock to Asia–Europe shipping capacity driven by the unilateral application of China's single-case terminal closure policy at one of the world's largest container ports,

entirely unrelated to any change in European trade demand.

Ocean Carriers Refusal to Accept Empty Container Returns Generates Disruptions, LA-LB (November–December 2021)

Technical details of the surcharge.— The surcharge was first announced on 3 November 2021 by MSC and extended in December 2021 by Maersk, both applied to the Ports of Los Angeles and Long Beach at a rate of $100 USD per container. The event is classified as an operational disruption.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
3 November 2021	1 November 2021	$100	MSC Customer Advisory
1 December 2021	29 December 2021	$100	Maersk Notice

Background.— The Ports of Los Angeles and Long Beach together constitute the largest container port complex in the Western Hemisphere, handling approximately 31 per cent of all United States ocean trade (Port of Los Angeles, 29 October 2021). Terminal operations at the complex depend on a continuous cycle in which drayage trucks collect loaded containers and return empty ones in a single trip; when ocean carriers stop accepting empty container returns, trucks are left stranded with boxes they cannot drop off, preventing them from collecting the next loaded container and causing imported cargo to accumulate on terminal indefinitely (FreightWaves, 25 October 2021).

Cause of the price surcharge.— The cause of both the November and December surcharges was the decision by ocean carriers to stop accepting the return of empty containers at their terminals, which brought the flow of cargo out of the port to a near-standstill. By 25 October 2021, a Harbor Trucking Association survey of 15 member companies found a combined 4,251 empty containers stuck on trucks with no terminal willing to take them back, with one carrier reporting it had been unable to return empties since 31 August (FreightWaves, 25 October 2021). This caused approximately 40 per cent of all containers at both ports to

dwell beyond nine days simultaneously, prompting the port authorities to impose a compulsory dwell fee on ocean carriers effective 1 November (Port of Los Angeles, 29 October 2021). The crisis deepened through December, when the number of vessels waiting offshore peaked at 94 and empty containers accumulated to over 90,000 units across the complex, forcing the extension of the dwell fee programme into the following month (Western Overseas, 24 December 2021). MSC and Maersk shifted the cost of the compulsory dwell fee directly onto consumers, a shock driven by the port authorities' direct response to carriers' own refusal to accept empty container returns, entirely unrelated to any change in US import demand.

War Disrupts Black Sea Trade (March 2022)

Technical details of the surcharge.— Announced on 2 March 2022 and implemented on 3 April 2022 by Maersk, the surcharge applied to cargo from and to all Black Sea ports at a rate of $1,100 USD per container (Maersk Notice). The event is classified as a war-related disruption.

Background.— The Black Sea is a semi-enclosed body of water connected to the Mediterranean solely through the Turkish Straits, making it the primary maritime gateway for the exports of Ukraine, Russia, Romania, and Bulgaria, with the Russian port of Novorossiysk and the Ukrainian ports of Odessa, Yuzhne, and Mariupol serving as the principal container and bulk cargo terminals for the region (Allianz). Marine war risk insurance for vessels operating in these waters is governed by a framework administered by the Lloyd's Joint War Committee, a body comprising underwriting representatives of the Lloyd's and London company markets that publishes a list of Areas of Perceived Enhanced Risk; when a region is designated, shipowners are required to notify their hull insurance underwriters before each transit and must pay additional premiums, known as breach additional premiums, to maintain coverage, with the level of those premiums set by individual underwriters based on their own risk assessment (Maritime Executive, 16 February 2022).

Cause of the price surcharge.— On 24 February 2022, Russia launched its full-scale invasion

of Ukraine, and on the same day a Russian missile struck the Turkish-owned bulk carrier Yasa Jupiter off Odessa, the first confirmed merchant vessel casualty of the conflict. On 25 February, two further civilian vessels were struck: the Japanese-owned bulk carrier Namura Queen was hit by a rocket at anchor off Yuzhne, and the Moldovan-flagged tanker Millennial Spirit was struck and set on fire twelve miles south of Odessa (Splash247, 26 February 2022). The three confirmed vessel losses in the first 48 hours of the conflict triggered an immediate response from the Lloyd's Joint War Committee, which expanded its war risk listed areas to cover the Sea of Azov and the northern Black Sea, with underwriters subsequently issuing cancellation notices and quoting breach additional premiums as high as 5–10 per cent of hull value per transit, a level at which many underwriters refused to write coverage at all (Lloyd's List, 4 March 2022). With Ukrainian ports closed by order of the country's maritime authorities and war risk insurance costs rendering calls to regional ports commercially unviable for most carriers, Maersk issued the Emergency Risk Surcharge on 2 March 2022 to offset the elevated insurance and operational costs of any residual service to Black Sea ports, a shock driven by the sudden onset of armed conflict and the consequent collapse of insurable access to the region, entirely unrelated to any change in trade demand.

Equipment Delays and Strike Hit Le Havre (April 2022)

Technical details of the surcharge.— Announced on 20 April 2022 and implemented on 19 April 2022 by Maersk, the surcharge applied to all Carrier Haulage and Multi-carrier bookings for cargo to and from the Terminal de Normandie MSC (TNMSC) at the Port of Le Havre at a rate of $135.4 USD per container (Maersk Advisory). The event is classified as a strike-related disruption.

Background.— The Terminal de Normandie MSC is one of the two principal deep-water container terminals at the Port of Le Havre, located in the Port 2000 basin and serving as a primary calling point for major carrier services on the Asia–Europe and transatlantic trade lanes. The terminal's operations depend on a fleet of straddle carriers to move containers

between ship-side and the storage yard; when multiple vessels are berthed simultaneously, the straddle carrier pool is shared across all active ships, reducing the throughput available to any individual vessel and slowing the gate movement of full containers to and from road hauliers.

Cause of the price surcharge.— The cause of the April surcharge was a convergence of three supply-side operational disruptions at the TNMSC terminal that developed in early April 2022. First, several shipping lines restructured their Asia–Europe service networks in April 2022, which redirected additional transshipment volumes through Le Havre at the expense of Antwerp, generating a higher number of vessel calls and container movements than the terminal had previously handled (Linerlytica Market Pulse, 21 February 2022). Second, the terminal had recently taken delivery of new straddle carrier equipment that was still undergoing calibration and software programming, reducing effective equipment availability. Third, dock workers announced an intermittent strike of undetermined duration starting in early April 2022, with recurring stoppages reducing operational shifts and directly cutting the number of container moves per day at a moment when the yard was already saturated (Balguerie). The combined effect drove average truck waiting times at the terminal gate to between four and six hours, causing inland transport suppliers to levy congestion fees against Maersk for all intermodal services at the terminal, which Maersk passed through as a surcharge on 20 April 2022, a shock driven by the simultaneous onset of a service network reconfiguration, equipment installation disruption, and industrial action, entirely unrelated to any change in French trade demand (Maersk Advisory).

India Dry Port Chaos After Price Control Regulation (May 2022)

Technical details of the surcharge.— Announced on 2 May 2022 and implemented on 1 June 2022 by Maersk, the surcharge applied to all Indian locations via Indian ports to the world at a rate of $24.71 USD per container (Maersk Notice). The event is classified as an operational disruption.

Background.— India's inland container logistics network relies on a system of Inland Container Depots, which serve as customs-cleared freight stations located away from seaports, connecting factories and industrial hubs to the major export gateways of Jawaharlal Nehru Port, Mundra, and Pipavav via road and rail (Maersk); the cost of moving containers between these depots and the seaports had long been elevated by two structural constraints, a chronic shortage of truck drivers and a railway network shared between passenger and freight trains, making carrier-managed inland haulage systematically more expensive in India than in comparable export economies (IRU).

Cause of the price surcharge.— The cause of the May 2022 surcharge was a sudden and concentrated shock to trucking costs across India triggered by the resumption of daily retail diesel price increases after a five-month politically mandated freeze. State-owned oil marketing companies had frozen retail fuel prices in early November 2021 at the onset of electioneering for legislative assembly elections in five states (BusinessToday). Following the conclusion of voting and the announcement of election results on 10 March 2022, daily price revisions were resumed, implementing nine consecutive increases in ten days by early April producing an immediate 25–30 per cent increase in full truck load freight rates, with industry executives warning that there was no way transporters could absorb the concentrated shock (Business Standard). Maersk issued the Dry Port Surcharge on 2 May 2022, a shock to inland container haulage capacity driven by the abrupt resumption of fuel price pass-through after a politically imposed freeze.

Strikes in Northern Europe (September 2022)

Technical details of the surcharge.— Announced and implemented on 7 September 2022 by Maersk, the surcharge applied to the Port of Le Havre Terminal de France (TDF) and the Port of Dunkirk at a rate of $122.5 USD per container (Maersk Notice 1, Notice 2). The event is classified as a strike-related disruption.

Background.— From July 2022, approximately 12,000 dockworkers represented by the Ver.di

union staged a series of warning strikes at the German North Sea ports of Hamburg, Bremerhaven, and Wilhelmshaven, the longest and most disruptive port labour action in Germany in decades, demanding a 14 per cent wage increase across 58 collective bargaining companies plus an inflation compensation bonus, shutting down terminals repeatedly and generating vessel queue and congestion spillover across the entire Hamburg–Le Havre range (Maritime Executive, 13 July 2022). A one-day national strike in Belgium simultaneously disrupted operations at Antwerp, and similar industrial action spread to France (Port Strategy, October 2022).

Cause of the price surcharge.— The immediate cause of the surcharge was strike action at all Le Havre terminals on 31 August 2022, which directly removed stevedoring capacity from a TDF terminal already running at critically high yard occupancy with systematic delays for all container entry and exit movements and average vessel discharge times of four days (Qualitairsea). The congestion at Dunkirk arose from the same regional overflow dynamic, with the port absorbing diverted vessel calls and container volumes that could not be processed at the saturated German and Belgian terminals (Maersk Advisory). Maersk issued the surcharge on 7 September 2022, a shock to landside container operations driven by strike-induced capacity removal, entirely unrelated to any change in French trade demand.

Diversion Over Strike Fears Overwhelms Port Houston (November 2022)

Technical details of the surcharge.— Announced on 15 November 2022 and implemented on 1 December 2022 by Maersk, the surcharge applied to the Port of Houston at a rate of $45 USD per container per day (Maersk Notice). The event is classified as an operational disruption.

Background.— From mid-2022, shippers began preemptively diverting cargo away from US West Coast ports in anticipation of disruption from the unresolved contract negotiations between the International Longshore and Warehouse Union and the Pacific Maritime Association, whose master agreement had expired in July 2022. The memory of the 2014–2015

ILWU dispute, which produced six months of significant port disruptions, made shippers unwilling to repeat the experience and prompted a structural shift of Asian import volumes to East and Gulf Coast gateways (The Loadstar, 24 October 2022). Houston absorbed a disproportionate share of this diverted traffic, with container volume rising 10 per cent year-on-year in July 2022, accelerating to 20 per cent in August, and reaching 26 per cent in September, pushing the port into the top five US container gateways for the first time, while import dwell times more than doubled from a baseline of three to four days to 6.5 days as the port's two terminals, Barbours Cut and Bayport, struggled to process volumes they had no capacity planning framework to absorb (Maritime Executive, 25 October 2022).

Cause of the price surcharge.— The surge produced a terminal operational crisis with no precedent in Houston's history as a container gateway: with long-dwelling loaded import containers continuing to accumulate on the docks and no sign of the volume pressure easing, the Port Commission of the Port of Houston Authority voted on 27 October 2022 to authorize the port's first-ever container dwell fee, a Sustained Import Dwell Fee of $45 per container per day from the eighth day after the expiration of free time, with a separate escalating Excessive Import Dwell Fee for containers dwelling beyond that threshold, as an emergency measure to restore terminal throughput at both Barbours Cut and Bayport (Maritime Executive, 1 November 2022). Maersk issued its advisory on 15 November 2022 to pass through this emergency port-imposed cost to its customers, a shock to Gulf Coast container capacity driven by the sudden and unplanned rerouting of US import volumes onto a port infrastructure that was never designed to serve as a primary Asia–US gateway, entirely unrelated to any change in Houston's underlying import demand.

ILWU Work Refusal on Pier 400 Automation, LA–LB (November 2022)

Technical details of the surcharge.— Announced on 22 November 2022 and implemented on 1 December 2022 by Maersk, the surcharge applied to APM Terminal Pier 400 at the Port of Los Angeles at a rate of $270 USD per container (Maersk Advisory). The event is classified

as an operational disruption.

Background.— APM Terminals Pier 400 is the largest marine terminal in the Port of Los Angeles complex, operating under a mandatory truck appointment system called TERMPoint through which trucking companies book import container pickup windows; for each labor shift, APM Terminals places dock labor orders based on confirmed appointment volumes, meaning that appointment no-shows translate directly into wasted labor deployment, idle equipment, and yard slots that cannot be reassigned to other operators, causing containers to accumulate on the terminal and progressively degrading throughput capacity for all users of the facility (APM Terminals). The causes of missed appointments cascade across the San Pedro Bay port complex: turn times of 90 to 120 minutes or longer at one terminal routinely prevent truck drivers from reaching their next terminal within any booking window, while warehouse congestion, chassis shortages, driver hours-of-service restrictions, and documentation delays further prevent truckers from honoring scheduled slots, each missed appointment deepening the yard congestion that in turn extends turn times and generates the next round of failures (ShipUWL, 21 December 2021).

Cause of the price surcharge.— The cause of the surcharge was a targeted labor action by ILWU dockworkers who, as of mid-September 2022, were refusing to work ships on the automated side of Pier 400, citing safety concerns, as part of a pattern of local disruptions tied to the stalled ILWU-PMA master contract negotiations; with the master contract having expired on 1 July 2022 without a replacement, the ports had no arbitration mechanism available to determine the legality of the action or compel a return to normal operations, leaving the refusal unresolved and reducing Pier 400's effective handling capacity through the autumn (Universal Cargo & Journal of Commerce). By late October 2022, ILWU-PMA negotiations had been formally suspended with no agreement expected before year end, meaning the labour pressure on Pier 400's operations continued unabated in the weeks immediately preceding the surcharge announcement (Universal Cargo). The resulting reduc-

tion in yard throughput extended vessel processing times and cascaded into truck scheduling chaos: by November 2022, missed and late truck appointments at Pier 400 had reached up to 70 per cent of all scheduled slots, wasting terminal labour ordered for those shifts and causing containers to accumulate in the yard beyond normal storage capacity (Maersk Advisory, 22 November 2022). Maersk issued the Late Gate Receiving Fee on 22 November 2022 to pass through the cost of those wasted terminal resources, a shock driven by the ILWU's targeted refusal to work Pier 400's automated facilities in the context of an unresolved and unarbitrable labour dispute, entirely unrelated to any change in US import demand.

India Dedicated Freight Corridor Expansion (January 2023)

Technical details of the surcharge.— Announced on 23 January 2023 and implemented on 20 February 2023 by Maersk, the surcharge reduction of average $250.85 USD per container applied to all Indian locations via Indian ports to the world (Maersk Notice). The event is classified as an operational disruption.

Background.— India's inland container logistics network relies on a system of Inland Container Depots, also known as dry ports, which serve as customs-cleared freight stations located away from seaports, connecting factories and industrial hubs to the major export gateways of Jawaharlal Nehru Port, Mundra, and Pipavav via road and rail (Maersk); the cost of moving containers between these depots and the seaports had long been elevated by two structural constraints, a chronic shortage of truck drivers and a railway network shared between passenger and freight trains, which imposed severe speed and capacity limitations on goods movement, making carrier-managed inland haulage systematically more expensive in India than in comparable export economies (IRU). To address the rail constraint, the Indian government initiated construction of two dedicated freight-only corridors, the Eastern Dedicated Freight Corridor from Ludhiana to Sonnagar (1,337 km) and the Western Dedicated Freight Corridor from Dadri to Jawaharlal Nehru Port (1,506 km), but the project had been subject to continuous delays since its original 2017 completion target, with deadlines

successively revised to March 2018, March 2020, December 2021, and June 2022, leaving the market with no reliable date for completion and every expectation of further slippage (The Tribune, 8 December 2022).

Cause of the price surcharge.— The cause of the January 2023 surcharge reduction was the unexpected commissioning of the Sujatpur–Karchana section (62 km) of the Eastern Dedicated Freight Corridor on 31 December 2022, which the market had not anticipated: as recently as 8 December 2022, just 23 days before commissioning, the Railway Minister had told Parliament only that remaining sections were *"targeted for completion in year 2023"*, with no indication any section was imminent, and a freight train trial on the section had only been completed a few days before the formal commissioning, meaning the final regulatory green signal had not yet been confirmed even on the eve of the year-end deadline (Cargo Insights, 2 January 2023). The successful trial run, a 53-wagon loaded coal train traversing the full 62.65 km including a 1 km bridge over the Yamuna River, unlocked the formal commissioning and extended seamless freight-only rail connectivity to 327 km between Bhimsen and Chheoki, directly reducing transit time and truck dependency for container cargo moving from inland freight stations to the ports of Jawaharlal Nehru and Mundra (DFC-CIL Milestones); by 31 January 2023, the Ministry of Railways confirmed that 1,724 km of dedicated freight corridors had been commissioned in total, from zero in 2014, with the imminent February 2023 commissioning of the New Bhimsen–New Bhaupur section set to extend seamless connectivity further to 752 km between Dadri, Khurja, Bhaupur, and Chunar (Press Information Bureau, 8 February 2023). Maersk issued the surcharge reduction on 23 January 2023, a supply-side cost relief driven by the surprise early commissioning of new dedicated freight rail capacity that materially reduced inland container haulage costs across the Indian subcontinent, entirely unrelated to any change in export demand.

Disruptions at Turkish Ports Following Earthquake (March–August 2023)

Technical details of the surcharges.— CMA CGM issued three successive congestion surcharges on Turkish port trade in the months following the 6 February 2023 earthquake. All three events are classified as weather-related disruptions.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
27 March 2023	27 March 2023	$100	CMA CGM Notice
7 April 2023	7 April 2023	$100	CMA CGM Notice
25 August 2023	1 September 2023	$100	CMA CGM Notice

Background.— On 6 February 2023, a 7.8-magnitude earthquake struck southern Turkey, causing catastrophic damage to the Port of Iskenderun — one of the two principal container hubs on Turkey's southeastern Mediterranean coast — which suffered severe structural damage, a fire triggered by the earthquake that destroyed over 3,600 containers, and an indefinite suspension of all operations (Ship Technology). Within hours of the earthquake, MAERSK and CMA CGM issued diversion advisories redirecting Iskenderun-bound cargo to alternative ports, with the Port of Mersin, the nearest fully operational container terminal, and Egyptian ports at Alexandria and Damietta absorbing the bulk of the displaced vessel traffic (Portcast, MAERSK, CMA CGM). The port partially reopened to commercial container operations after 8 April 2023, but the earthquake's destruction of the terminal's fixed gantry cranes and quay infrastructure was permanent: the port could only resume operations at 45 per cent of its pre-earthquake throughput capacity, confined exclusively to container services with degraded equipment (Digital Logistics Capacity Assessments; Ports Europe).

Cause of the surcharges.— The sudden closure of Iskenderun compressed the entire southeastern Turkish container trade onto Mersin's terminal infrastructure, producing a sustained surge in vessel arrivals and yard occupancy that overwhelmed the port's capacity and progressively degraded landside truck turnaround times as regional road networks simultaneously absorbed diverted haulage traffic (More Than Shipping). The 27 March surcharge addressed the export congestion from Mersin across all outbound trades; the 7 April surcharge extended

the mechanism to inbound Asian trades as vessel bunching on the eastbound leg into Mersin compounded the existing yard saturation (Hellenic Shipping News). The third and final phase of the surcharge sequence arose when Iskenderun itself reopened with permanently halved handling capacity: by August 2023, the structural mismatch between resuming vessel arrival volumes and the port's degraded infrastructure had produced average congestion of ten days per vessel call, a level CMA CGM described as beyond its control and not expected to improve, prompting the carrier on 23 August 2023 to suspend all direct Iskenderun calls on its NC LEVANT EXPRESS North Europe service and reroute cargo via transhipment through Beirut, before issuing the 25 August surcharge to recover the operational costs of the congestion (American Journal of Transportation, 23 August 2023). All three surcharges were shocks driven by the earthquake's destruction of Iskenderun's handling capacity, entirely unrelated to any change in Turkish trade demand.

Panama Canal Congestion Due to El Niño Drought (April–July 2023)

Technical details of the surcharges.— CMA CGM and Maersk issued successive surcharges on cargo transiting the Panama Canal as drought restrictions progressively tightened through 2023. Both events are classified as weather-related disruptions.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
6 April 2023	1 June 2023	\$600	CMA CGM Notice
7 July 2023	3 August 2023	\$240	Maersk Notice

Background.— The Panama Canal relies on freshwater from Gatun Lake, fed by rainfall in the canal watershed, to operate its lock system; when lake levels fall below operational thresholds, the Canal Authority is compelled to reduce the maximum draft permitted for transiting vessels and to limit the number of daily transits, forcing carriers either to lighten vessel loads, take on less cargo, or reroute entirely via the Suez Canal or Cape of Good Hope (Woodwell Climate Research Center). In 2023, a severe drought intensified by the El Niño phenomenon produced a prolonged water shortage, with water levels in Gatun Lake falling

to their lowest recorded since at least 1965, a hydrological extreme entirely outside historical operational experience for the Canal Authority (World Weather Attribution).

Cause of the surcharges.— The April surcharge was triggered by an acute deterioration in hydrological conditions in the Panama Canal watershed. In the first eleven days of April 2023, precipitation across the watershed fell to approximately 70% below the historical average, with river inflows declining to 80% below average; the Canal Authority responded by issuing emergency advisories announcing a cascade of Neopanamax draft reductions, lowering the maximum authorised draft in four successive steps from 47.5 feet on 19 April to 46.0 feet on 12 May (gCaptain, 19 April 2023). The restrictions compelled carriers on the Northeast Asia–East Coast Central America trade to lighten vessel loads to remain within the incoming draft limits, directly raising per-unit operating costs and prompting CMA CGM to issue its surcharge on 6 April. Despite the nominal onset of Panama's rainy season, conditions deteriorated further through the second quarter: by early July 2023, Gatun Lake had fallen to a four-year low of 79.7 feet against a seasonal norm of 84 feet, with projections indicating a further decline to 79.5 feet by August; the Canal Authority confirmed that daily transit slots had been reduced to 30–31 vessels against a normal allocation of 36–37, with 59 vessels already queuing at each end of the waterway. On 6 July, the Canal Administrator convened an emergency session to consider structural interventions, including the sourcing of water from catchments outside the existing watershed (gCaptain, 2 July 2023). Maersk issued the 7 July surcharge the following day, extending the cost-pass-through to all cargo transiting the Canal regardless of trade direction as the crisis entered a phase with no prospect of near-term hydrological recovery. Both surcharges were shocks driven by an exogenous hydro-meteorological event that reduced the physical throughput capacity of the Canal, entirely unrelated to any change in trade demand on the affected routes.

Further Restrictions Panama Canal Due to El Niño Drought (November 2023)

Technical details of the surcharge.— Announced on 21 November 2023 by CMA CGM and on 23 November 2023 by MSC, and implemented from 1 January 2024 and 15 December 2023 respectively, the surcharge applied to all cargo transiting the Panama Canal at a rate of $298.5 USD per container (CMA CGM Notice, MSC Notice). The event is classified as a weather-related disruption.

Background.— The Panama Canal relies on freshwater from Gatun Lake to operate its lock system; when lake levels fall below operational thresholds the Canal Authority reduces the maximum draft permitted for transiting vessels and limits the number of daily transits, forcing carriers to lighten loads or reroute around Cape Horn. The 2023 El Niño drought had been driving progressive restrictions since April, but conditions continued to deteriorate through the northern hemisphere autumn rather than recovering as the rainy season advanced: Gatun Lake water levels hovered near 80 feet between June and November 2023, far below the typical 88 feet for the end of the rainy season, and the Canal Authority projected levels to remain below 81 feet at least through March 2024 (Bureau of Transportation Statistics).

Cause of the surcharge.— October 2023 recorded the lowest precipitation since 1950, at 41 per cent below historical norms, preventing any seasonal recovery in Gatun Lake; on 31 October 2023, the Canal Authority announced a cascade of transit reductions descending from 25 slots per day in early November, to 24 from 7 November, to 22 in December, to 20 in January 2024, and to 18 from February 2024 onwards, the most severe climate-related transit restriction since the expanded Neopanamax locks opened in 2016 (Seatrade Maritime News). By 7 November, booking slots had already been reduced to 24 per day, down 33 per cent from the pre-drought norm of 36–38, with 59 vessels queuing for transit slots as waiting times on the Pacific side climbed to over two days (Supply Chain Dive). On 24 November 2023, the Canal Authority formalised the booking system restrictions for the entire period, confirming

that the Transit Reservation system was the only remaining mechanism available to carriers to guarantee a transit date (BTS, citing ACP Advisory A-50-2023). CMA CGM and MSC issued the surcharge on 23 November 2023, a shock driven by the unprecedented severity of the El Niño-induced drought on Canal throughput capacity, entirely unrelated to any change in trade demand.

Red Sea Conflict Forces Cape of Good Hope Rerouting (December 2023)

Technical details of the surcharge.— Announced on 20 December 2023 and implemented on 1 January 2024 by MSC, the surcharge applied to all cargo on the Europe and Mediterranean to and from Asia trade at a rate of $891.67 USD per container (MSC Notice). The event is classified as a war-related disruption.

Background.— The Red Sea and Bab el-Mandeb Strait constitute the sole maritime corridor linking the Indian Ocean to the Suez Canal, through which approximately 30 per cent of global container trade between Asia and Europe transits; any closure of the strait forces carriers onto the alternative route around the Cape of Good Hope at the southern tip of Africa, adding approximately 11,000 nautical miles and two additional weeks of transit time to every affected voyage (Maritime Economics & Logistics). From October 2023, Houthi forces in Yemen began firing missiles and drones in solidarity with Gaza, initially targeting Israeli military assets; on 19 November 2023 the Houthis seized the car carrier Galaxy Leader, the first commercial vessel captured, and subsequently escalated to attacks on merchant vessels transiting the lower Red Sea and the Bab el-Mandeb using drones, missiles, and armed speedboats, initially claiming to target only vessels with links to Israel (UN Security Council Report, 2 January 2024).

Cause of the surcharge.— On 9 December 2023 the Houthis publicly expanded their target list to all commercial vessels heading to Israeli ports regardless of flag or ownership, transforming the threat from a targeted political action into an indiscriminate blockade of one of the world's most important container shipping corridors (UN Security Council Report,

2 January 2024). On 14 December a near-miss attack on the Maersk Gibraltar prompted Maersk to suspend all Red Sea and Gulf of Aden transits on 15 December, while on 18 December BP became the first oil major to suspend all tanker transits through the corridor and the United States announced the establishment of Operation Prosperity Guardian, a multinational naval protection task force (Maersk Advisory, 15 December 2023; Fortune, 4 January 2024). By 19 December 2023, with attacks continuing despite the naval operation, Maersk confirmed that all vessels previously scheduled to transit the Red Sea were being permanently rerouted via the Cape of Good Hope, with approximately 20 vessels already stopped en route (CNBC, 19 December 2023). MSC issued the Contingency Adjustment Surcharge on 20 December 2023, a shock driven by the sudden and complete closure of the world's principal Asia–Europe container corridor by an exogenous military conflict, entirely unrelated to any change in Asia–Europe trade demand.

Low Water Level, St. Laurent Seaway (October–November 2024)

Technical details of the surcharges.— Multiple carriers issued successive low water surcharges on cargo moving to the Port of Montreal across October and November 2024. Both events are classified as weather-related disruptions.

Announcement	Implementation	Rate (USD/FEU)	Carrier Notice
21 October 2024	1 November 2024	$400	CMA CGM, Maersk, MSC
4 November 2024	1 December 2024	$300	CMA CGM

Background.— The navigation channel on the lower St. Laurent River between Québec City and the Port of Montreal operates within tight physical tolerances: the maximum draft permitted for vessels transiting the channel varies between 8.00 and 8.08 metres depending on prevailing water levels, which are primarily determined by outflows from Lake Ontario and by seasonal precipitation across the Great Lakes watershed (Clear Seas). Because carriers must load vessels several weeks before arrival at Montreal in order to comply with the forecast draft at arrival, any reduction in channel depth translates directly and mechanically

into a reduction in cargo carried per sailing, forcing vessels to depart partially empty to ensure safe transit through the shallowest sections of the channel. The lower St. Laurent had experienced episodic low-water conditions requiring carrier surcharges on multiple prior occasions documented in this analysis.

Cause of the surcharge.— Water levels on the St. Laurent had been declining sharply since late summer 2024; by 4 October, the Canadian Coast Guard's forecasts were already predicting no near-term recovery, prompting the first wave of carriers to announce low water surcharges citing the CCG forecasts and the resulting reduction in vessel carrying capacity (Kuehne+Nagel, 4 October 2024; MSC, October 2024). Rather than recovering as the seasonal low-water period passed, the situation deteriorated further: the CCG bulletin issued on the morning of 4 November 2024, the same day CMA CGM and Maersk announced their surcharge, forecast water levels at Montreal to be exceptionally low until December (Canadian Coast Guard, St. Lawrence Ship Channel Water Level Forecasts Bulletin, 4 November 2024), requiring carriers to further reduce cargo loads below the already reduced levels they had been accepting since October. All surcharges in this episode were shocks driven by adverse hydrological conditions reducing the physical cargo-carrying capacity of vessels on the Montreal trade lane, entirely unrelated to any change in Canadian import demand.

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